

CARBON & ITS COMPOUND

- **Carbon**

- Carbon word derived from Latin word 'carbo'.

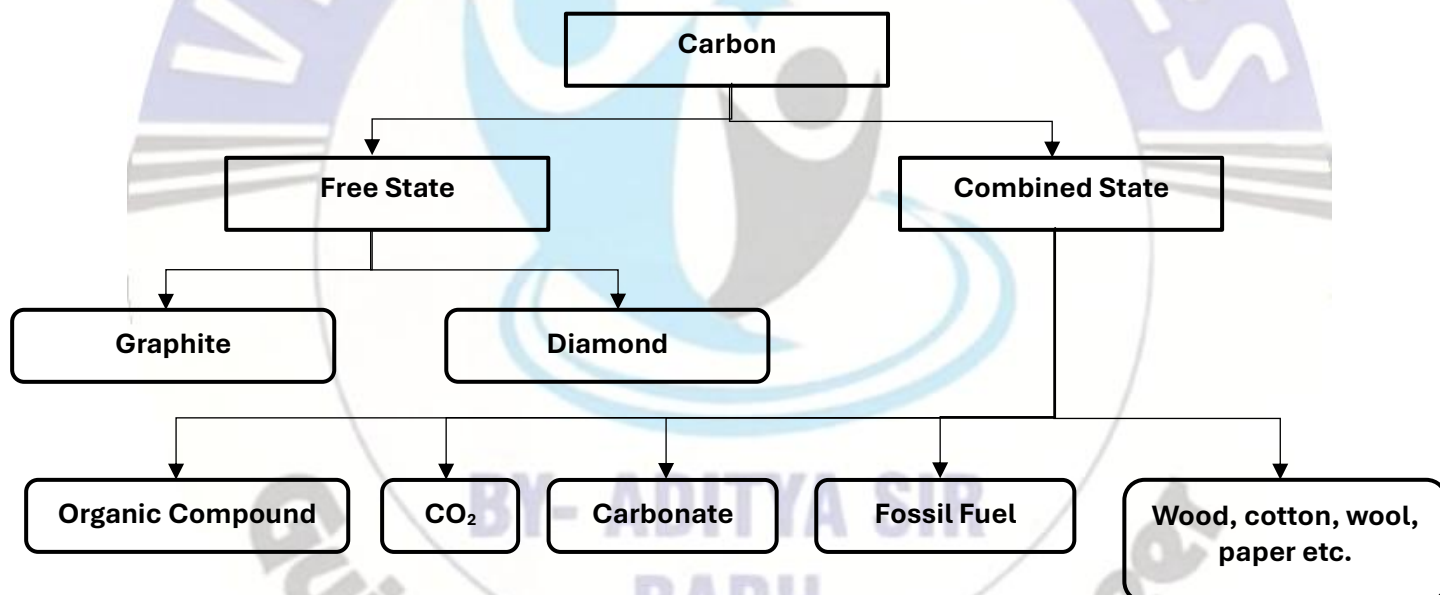
→ means black colour

Symbol → C [Non – metal]

${}_6\text{C}^{12}$ → Mass number

→ Atomic number

- The amount of carbon is 0.02% in the earth crust.
- The amount of carbon is 0.03% in the atmosphere.
- It contains 4 electrons in the outermost shell (valence electron). It makes only covalent bond i.e, sharing of electrons.
- It has no free electrons.



- Diamond and graphite are allotropes of carbon.
- Buckminsterfullerene is a new allotrope of carbon which was discovered recently.

- **Allotropes**

- The different forms of an element which are found in nature are called allotropes of that element.

Example:-

- ✓ Carbon has three allotropes.
 - i) Diamond.
 - ii) Graphite.
 - iii) Buckminsterfullerene.

✓ Hydrogen has three allotropes.

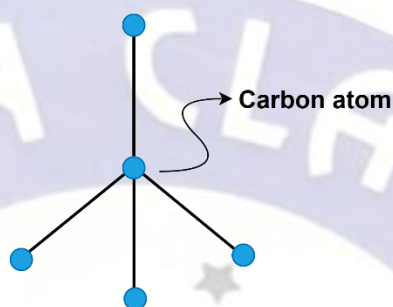
- i) $1\text{H}^1 \rightarrow$ Protium
- ii) $1\text{H}^2 \rightarrow$ Deuterium
- iii) $1\text{H}^3 \rightarrow$ Tritium

• **Diamond**

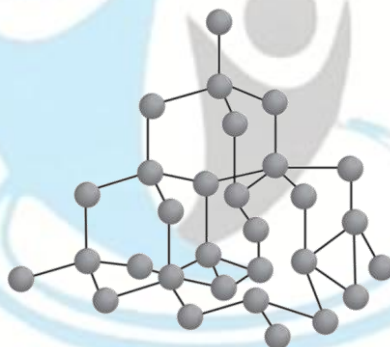
➤ Diamond is the purest form of carbon.

OR

Diamond is allotropes of carbon.



Regular Tetrahedral



Structure of Diamond

• **Properties of Diamond**

- i) Diamond is a transparent and extremely hard substance.
- ii) It is quite heavy.
- iii) It is non conductor of electricity because diamond has no free electron.
- iv) It is insoluble in water.
- v) It is unreactive or inert.
- vi) It is transparent for x – ray and opaque for duplicate diamond.
- vii) Its refractive index is very high (2.42), due to this reason diamond has extra ordinary brilliance.
- viii) Its density is 3500 kg/m^3 (High) and relative density is 3.52.
- ix) Each carbon atom in diamond is joined to 4 other carbon atom.

Special Note:

- Diamond is the hardest natural substance in the universe.

- **Uses of Diamond**

- i) Black Diamond is known as 'Bort' which is used in cutting glass, marbles, polishing the stones, grinding and drilling of stones and other hard substances.
- ii) It is used for making jewellery/ornaments (transparent diamond).
- iii) It is used by eye surgeon as a tool.

- **Artificial Diamond**

- When pure carbon is subjected to very high pressure and temperature for a long time then we obtained artificial diamond. It is also called synthetic diamond.

- **Presence of Diamond in India**

City	State
Golkunda	Hyderabad
Anantpur	Andhra Pradesh
Bellari	Karnataka
Panna	Madhya Pradesh

- **Mines of diamond in the World**

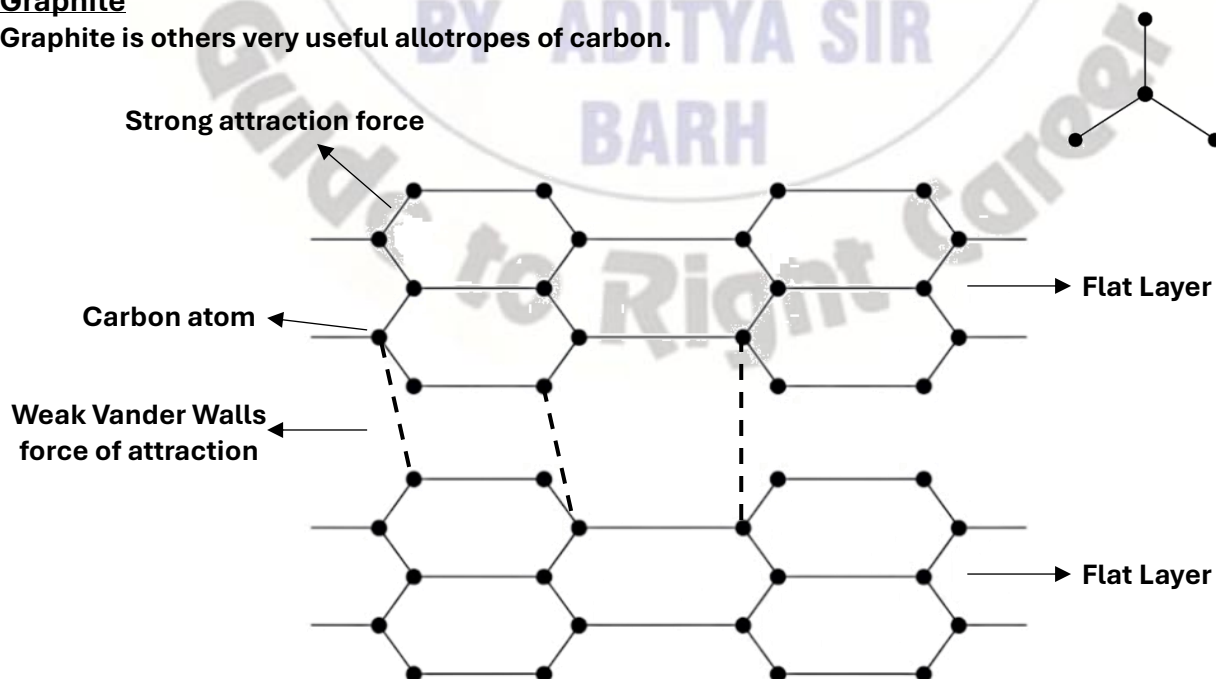
- i) South Africa (King of Diamond).
- ii) Brazil.
- iii) Austria.
- iv) America.

Types of Diamond

Diamond	Carats
Hope	44.5
Pit	136.5
Kohinoor	186
Cullinan (Rare present)	303.2

- **Graphite**

- Graphite is one of the very useful allotropes of carbon.



Structure of Graphite

- **Properties of Graphite**

- Graphite is a greyish black opaque substance.
- It is lighter than diamond.
- It is conductor of electricity.
- It is active than diamond.
- It is soft and slippery to touch.
- Its density is 2200 kg/m^3 and its relative density 2.2.
- Each carbon atom in graphite is joined to only three carbon atoms, thus each carbon has 1 free electron. Due to which it conduct electricity.

- **Uses of graphite**

- It is used in the form of electrode in Dry cell.
- It is used as electric arc in motor, dynamo etc.
- It is used for making pencil lid and black paints.
- It is used as lubricant for a machine.

Note:

- When we heat the graphite in the presence of very high pressure and temperature for a long time, then it changes in diamond.

- **Buckminsterfullerine**

- It is a new allotropes of carbon which is discovered recently.
- It has 60 carbon atoms joined together to form spherical molecule due to which it is also called Bucky ball.
- It is simply called “fullerence”.
- It is neither soft nor hard.
- It is dark solid at room temperature.

- **Source of Organic Compound**

- Microorganism plant, minerals etc.
- Carbon compound / organic compounds having low melting and boiling point.

ORGANIC CHEMISTRY

- **Introduction**

- All the compound which were directly or indirectly synthesized by living organisms are called organic compound and branch of chemistry which deals with these compound is called organic chemistry. In 1675, A French, chemist Nicolas Lemery.

D.O.B = 17 Nov 1645

D.O.D = 19 June 1715

Published his “coursede Chymine”.

- **In which he classified the substance into three parts.**

Mineral Kingdom → Inorganic Chemistry

Plant / Vegetable Kingdom

Animal Kingdom

} Organic Chemistry

In 1785 Birth period of Chemistry
According to Antonie D Lavoiser

- Antonie Laurent D Lavoiser

(A French Chemist
1st nobel prize man / Nobel man)

D.O.B → 26 Aug 1743

D.O.D → 8 May 1794

According to Lavoiser,

Compounds which were obtained directly or indirectly synthesized by living organism always contain carbon and Hydrogen.

Compound which are synthesized by living organism are called organic compound.

For example:-



Compound which are not synthesized by living organism are called inorganic compounds.

For example:-

✓ H_2O , HNO_3 , H_2SO_4 , HClO_4 etc.

In 1815

According to Jon Jacob Berzelius (An Swedish Chemist)

D.O.B → 20 Aug 1779

D.O.D → 7 Aug 1848

J.J Berzelius gives "Vital force Theory".

According to Berzelius, some vital force / life force are necessary to produce organic compounds, This theory is known as vital force theory.

No. of organic compounds is directly or indirectly synthesized in lab.

For example:-

Name	Vital Force	Organic compound
Red ant	Forma	Formic acid
Lemon	Citrus	Citric acid
Apple	Malus	Malic acid
Sour	Lactogen	Lactic acid
Urine	Human	Urea (NH_2CONH_2), Uric acid

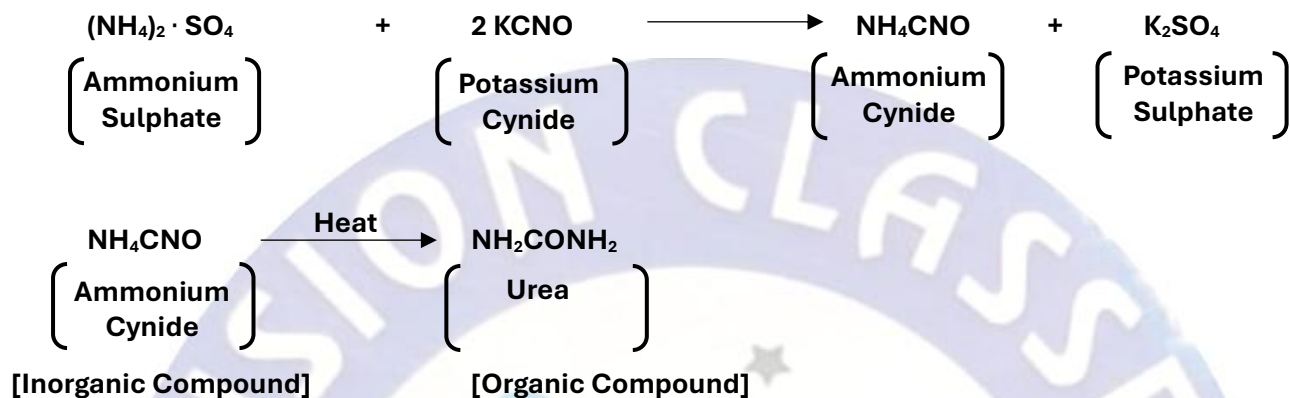
In 1828

A German Scientist → Friedrich Wohler

D.O.B → 13 Aug 1800

D.O.D → 23 Sep 1862

Wohler rejected the Berzelius vital force theory, he produces organic compounds by inorganic compounds.



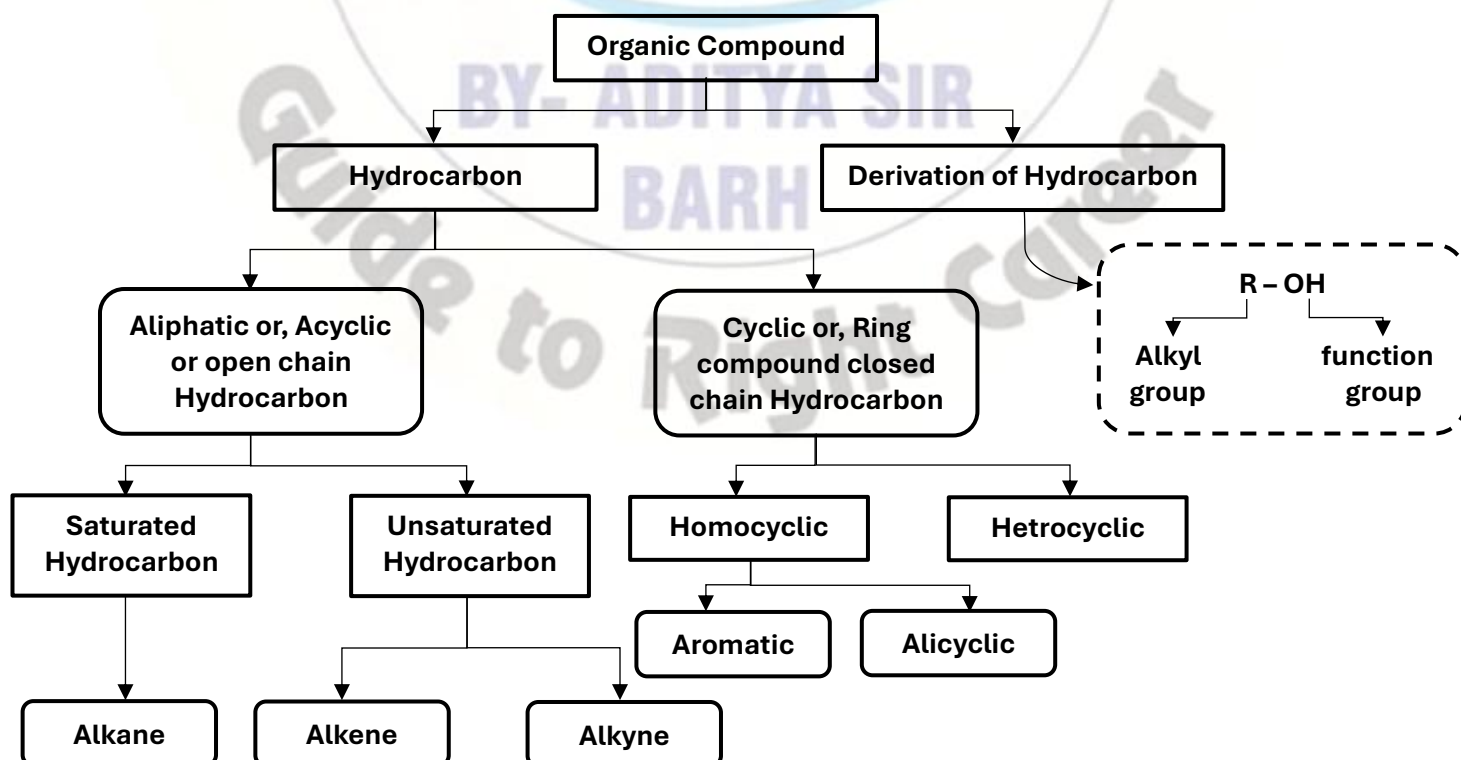
According to Modern Definition

Organic Chemistry is the branch of experimental science which deals with Hydro-carbon and derivative of hydro carbon.

i.e, Hydrocarbon → Hydrogen + Carbon

Special Note:

- Many more examples of organic compound urea, protein, fats, uric acid, carbohydrates (sugar, starch), oil food items etc.



- **Hydrocarbon**

- Those compound which have Hydrogen as well as carbon are called Hydrocarbon.

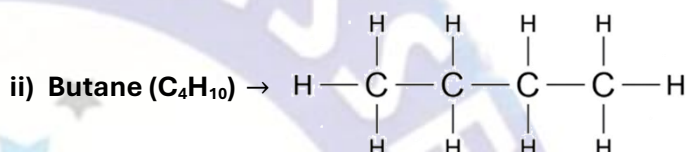
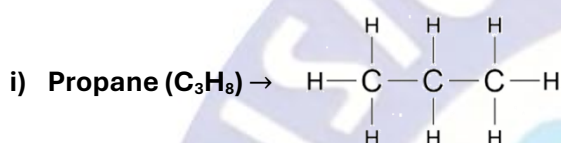
For example:-

- ✓ CH_4 , C_2H_4 , C_3H_8 etc.

- Hydrogen must be attached with carbon.
- Hydrocarbon are the important source of energy.

- **Aliphatic / Acyclic / Open chain Hydrocarbon**

- Those hydrocarbons in which carbon having a long chain. These compounds are called acyclic compound.



Note:

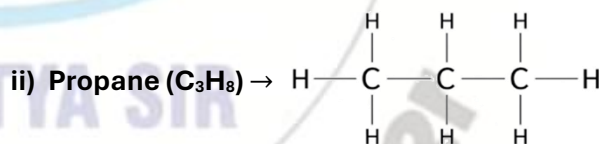
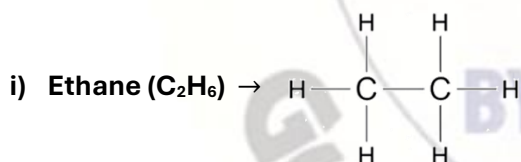
- Branch & Unbranch are also possible.

- **Saturated Hydrogen Carbon**

- These Hydrocarbons in which carbon – carbon and carbon – hydrogen are connected with only single bond are called saturated hydrocarbon.

For example:-

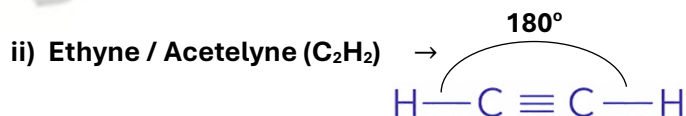
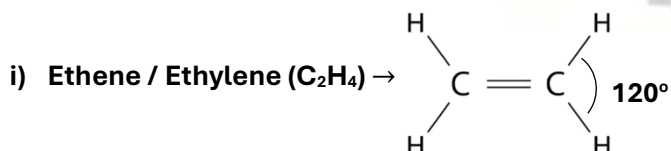
Alkane group



- **Unsaturated Hydrocarbon**

- Those hydrocarbon in which carbon – carbon atoms connected with multiple bond (Double or Triple bond).

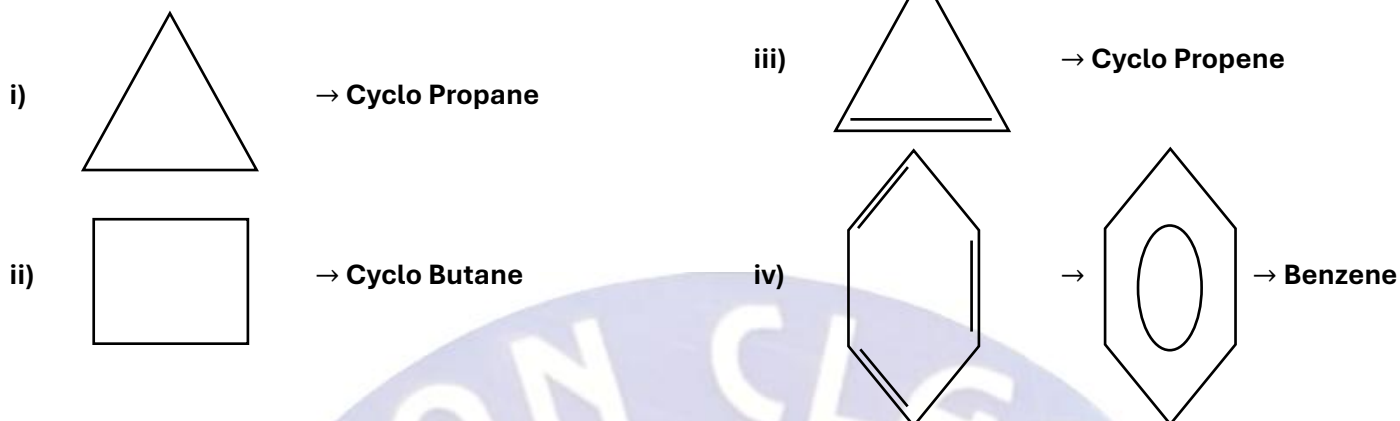
For example:-



- **Cyclic / Ring compound / Closed chain Hydrocarbon**

- Those hydrocarbon in which carbon having a ring chain. These compounds are called cyclic compound.

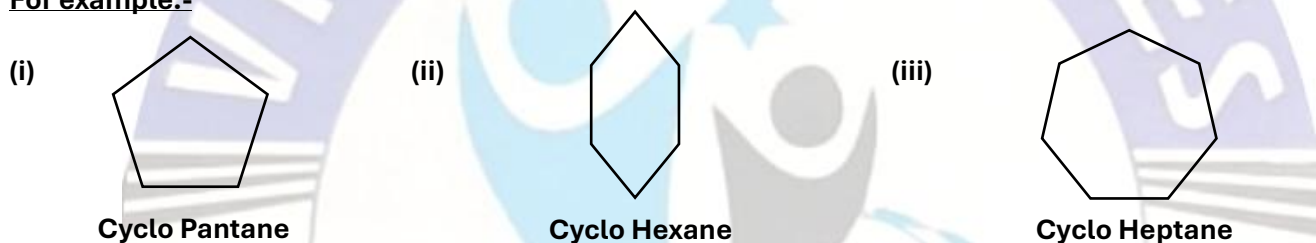
For example:-



- Homocyclic Compound

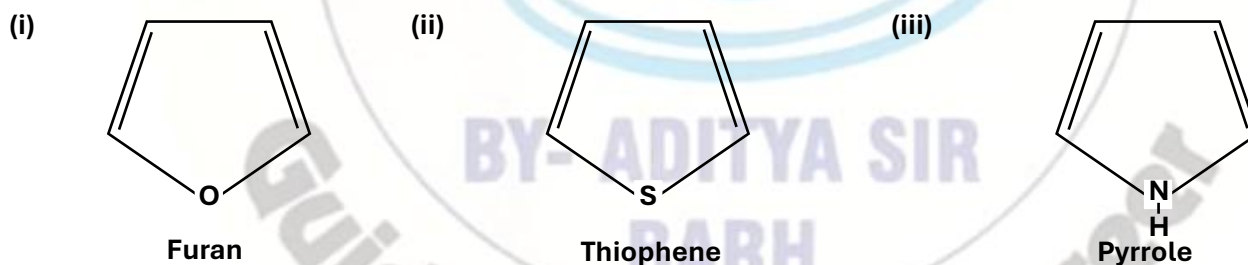
- Those organic compound in which bonds are connected by only carbon atom in the complete ring.

For example:-



- Hetrocyclic compounds

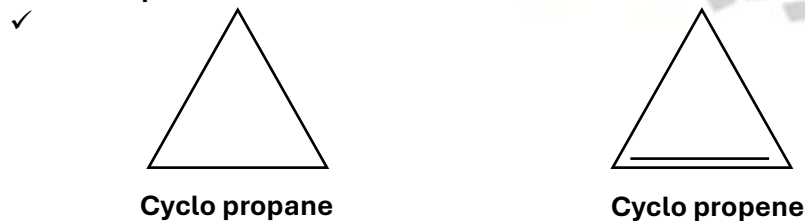
- There are cyclic compounds having ring or rings build up of more than 1 kind of atom.



- Alicyclic Compounds

- These are the compounds the properties like as aliphatic compound. These may be saturated or unsaturated.

For example:-



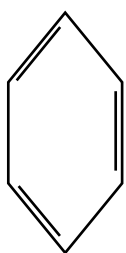
- Aromatic compounds

- These compounds consists of at list one **Benzene** ring called aromatic compound.

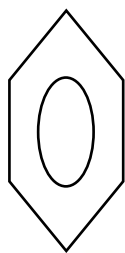
Benzene

A six numbered carbon cyclic ring having alternate double bond is called benzene ring.

i.e,



Benzene



Benzene

For example:-

CH₃



Toluene

OH



Finol (C₆H₅OH)

- **Nomenclature**

- The system of naming of organic compounds is called nomenclature.

No. of C – atom	Prefix
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex
7	Hept
8	Oct
9	Non
10	Dec
11	Undec
12	Dodec
13	Tridec
14	Tetradec
15	Pentadec
20	Eicos
21	Henicos
22	Decos
30	Tricot
40	Tetracot
100	Hect

- The nomenclature of organic compounds is written in two forms:

i) Common name (famous name in some particular area)

ii) I.U.P.A.C name

→ International Union of pure and applied chemistry

• **Saturated Hydrocarbon**

○ **Alkane**

Suffix → ane [C – C, C – H]
 General Formula → C_nH_{2n+2} , $n = N$

	No. of C – atom	M.F	I.U.P.A.C name	Structural formula
(i)	1	$C_1H_{2 \times 1 + 2} = CH_4$	Meth + ane → Methane Prefix Suffix	<pre> H H — C — H H </pre>
(ii)	2	C_2H_6	Eth + ane → Ethane	<pre> H H H — C — C — H H H </pre>
(iii)	3	C_3H_8	Propane	<pre> H H H H — C — C — C — H H H H </pre>
(iv)	4	C_4H_{10}	Butane	<pre> H H H H H — C — C — C — C — H H H H H </pre>
(v)	5	C_5H_{12}	Pentane	<pre> H H H H H H — C — C — C — C — C — H H H H H H </pre>
(vi)	6	C_6H_{14}	Hexane	<pre> H H H H H H H — C — C — C — C — C — C — H H H H H H H </pre>
(vii)	7	C_7H_{16}	Heptane	<pre> H H H H H H H H — C — C — C — C — C — C — C — H H H H H H H H </pre>
(viii)	8	C_8H_{18}	Octane	<pre> H H H H H H H H H — C — C — C — C — C — C — C — C — H H H H H H H H H </pre>
(ix)	9	C_9H_{20}	Nonane	<pre> H H H H H H H H H H — C — C — C — C — C — C — C — C — C — H H H H H H H H H H </pre>
(x)	10	$C_{10}H_{22}$	Decane	<pre> H H H H H H H H H H H — C — C — C — C — C — C — C — C — C — C — H H H H H H H H H H H </pre>

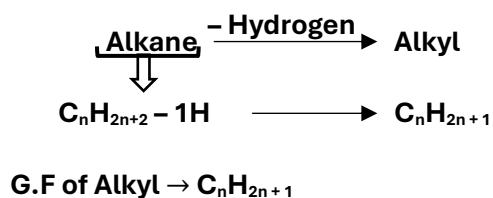
○ **Alkyl**

→ It is nothing

➤ It can't be prepared in lab.

➤ It can be derived by Alkane.

i.e, Alkyl group is prepared by removal of 1 hydrogen atom from alkane.



Suffix $\rightarrow$ yl
 General Formula $\rightarrow \text{C}_n\text{H}_{2n+1}$

➤ Alkyl group is denoted by 'R-'.

Free bond i.e, any atom / compound come to it and combine with it.

	No. of C – atom	M.F	I.U.P.A.C name	Structural formula
(i)	1	$\text{C}_1\text{H}_{2 \times 1 + 1} = \text{CH}_3$	Meth + yl $\rightarrow$ Methyl	<pre> H — C — H H </pre> Free bond
(ii)	2	$\text{C}_2\text{H}_{2 \times 1 + 1} = \text{C}_2\text{H}_5$	Eth + yl $\rightarrow$ Ethyl	<pre> H H H — C — C — H H </pre>
(iii)	3	C_3H_7	Propyl	<pre> H H H H — C — C — C — H H H </pre>
(iv)	4	C_4H_9	Butyl	<pre> H H H H H — C — C — C — C — H H H H </pre>
(v)	5	C_5H_{11}	Pentyl	<pre> H H H H H H — C — C — C — C — C — H H H H H </pre>
(vi)	6	C_6H_{13}	Hexyl	<pre> H H H H H H H — C — C — C — C — C — C — H H H H H H </pre>
(vii)	7	C_7H_{15}	Heptyl	<pre> H H H H H H H H — C — C — C — C — C — C — C — H H H H H H H </pre>
(viii)	8	C_8H_{17}	Octyl	<pre> H H H H H H H H H — C — C — C — C — C — C — C — C — H H H H H H H H </pre>
(ix)	9	C_9H_{19}	Nonyl	<pre> H H H H H H H H H H — C — C — C — C — C — C — C — C — C — H H H H H H H H H </pre>
(x)	10	$\text{C}_{10}\text{H}_{21}$	Decyl	<pre> H H H H H H H H H H H — C — C — C — C — C — C — C — C — C — C — H H H H H H H H H H </pre>

Note:

i) Free bond takes place either first carbon atom or last carbon atom.

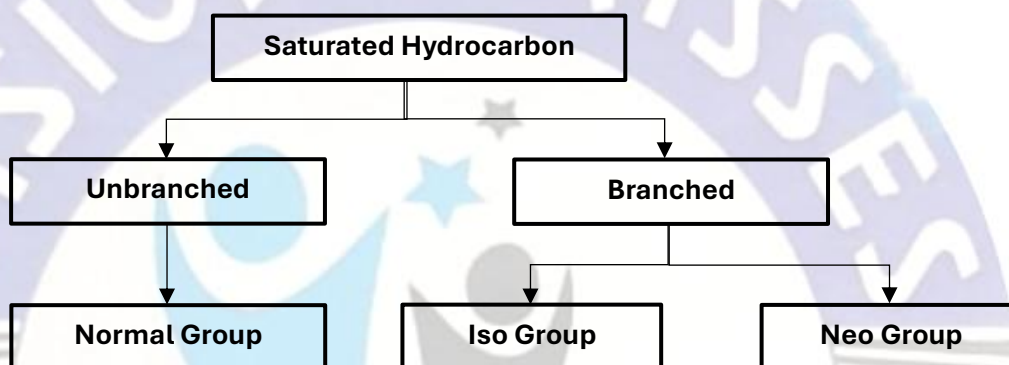
ii) Methane (CH_4) $\xrightarrow{-\text{H}}$ Methyl (CH_3-)

Ethane (C_2H_6) $\xrightarrow{-\text{H}}$ Ethyl (C_2H_5-)

Propane (C_3H_8) $\xrightarrow{-\text{H}}$ Propyl (C_3H_7-)

i.e, Alkyl is derivation of Hydrocarbon.

- Branched and Unbranched are also possible in Saturated Hydrocarbon.**



➤ Normal, Iso and Neo are common names.

- Normal Group**
- The straight chain of hydrocarbon is called normal group.
- Minimum number of carbon atom for normal is 4.

For example:-

Alkane	Alkyl
C_4H_{10} <pre> H H H H H-C-C-C-C-H H H H H </pre> C.N → Normal butane / n – butane I.U.P.A.C name → Butane	C_4H_9 <pre> H H H H H-C-C-C-C- H H H H </pre> C.N → Normal Butyl / n – butyl I.U.P.A.C → Butyl
C_5H_{12} <pre> H H H H H H-C-C-C-C-C-H H H H H H </pre> C.N → Normal Pentane / n – pentane I.U.P.A.C → Pentane	C_5H_{11} <pre> H H H H H H-C-C-C-C-C- H H H H H </pre> C.N → Normal Pentyl / n – pentyl I.U.P.A.C → pentyl

• **Iso Group**

- When one methyl group is attached with second last carbon atom of a hydrocarbon chain, called iso group.
- Minimum number of carbon for Iso is 4.

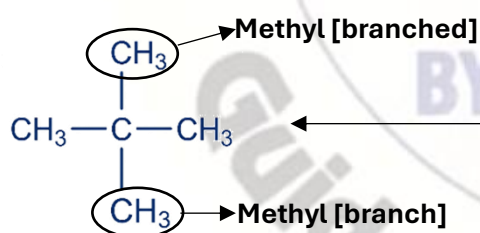
Alkane	Alkyl
C₄H₁₀ CH₃-CH-CH₃ CH₃ → methyl → Branch OR $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & \\ \text{H} & & \text{H} \\ & & \\ & \text{H}-\text{C}-\text{H} \\ & & \\ & \text{H} & \end{array}$ C.N → Iso butane I.U.P.A.C name → 2 - methyl propane	C₄H₉ CH₃-CH-CH₂- CH₃ C.N → Iso butyl I.U.P.A.C → 2 - methyl propyl C₅H₁₂ $\begin{array}{c} 1 & 2 & 3 & 4 \\ \text{CH}_3 & -\text{CH} & -\text{CH}_2 & -\text{CH}_3 \\ & & & \\ & \text{CH}_3 & & \end{array}$ C.N → Iso pentane I.U.P.A.C name → 2 - methyl butane

• **Neo Group**

- When two methyl groups attached with second last carbon atom of a straight chain hydrocarbon called neo group.
- Minimum number of carbon atom for neo is 5.

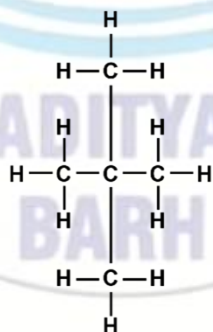
For example:-

✓ **C₅H₁₂**

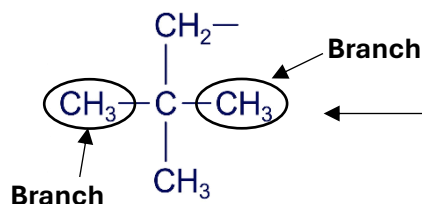


C.N → Neo Pentane

I.U.P.A.C name → 2, 2 - dimethyl propane

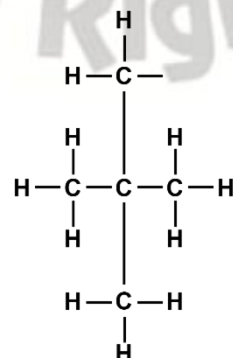


✓ **C₅H₁₁**



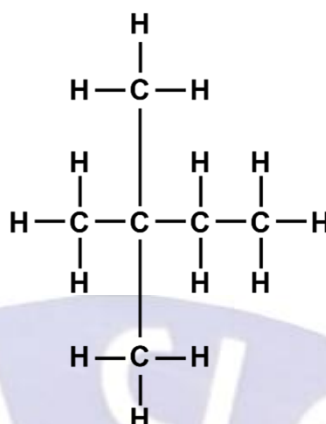
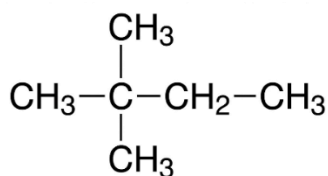
C.N → Neo pentyl

I.U.P.A.C name → 2, 2 - Dimethyl propyl



→ branch

✓ C_6H_{14}



C.N → Neo hexane

I.U.P.A.C name → 2, 2 - dimethyl butane

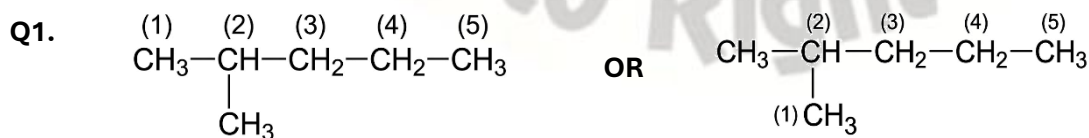
- In the process of I.U.P.A.C name
 - Commas (,) between digit & digit.
 - Hyphen (-) between digit & word.
 - Nothing between word & word.
- Special Greek prefix for no. of group

No. of groups	Prefix
2	Di
3	Tri
4	Tetra
5	Penta
6	Hexa
7	Hepta
8	Octa

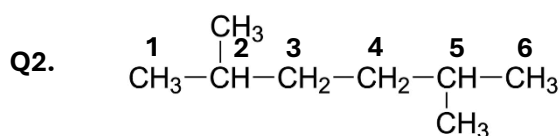
Note:

- Counting of carbon atoms must be started towards either free bond or nearest branch or functional group.
- We select only longest chain of hydrocarbon.

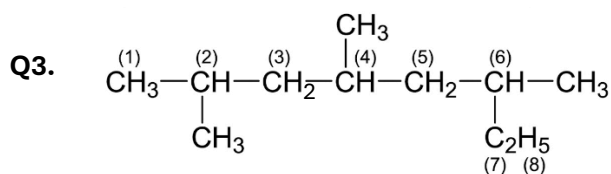
- Write I.U.P.A.C name



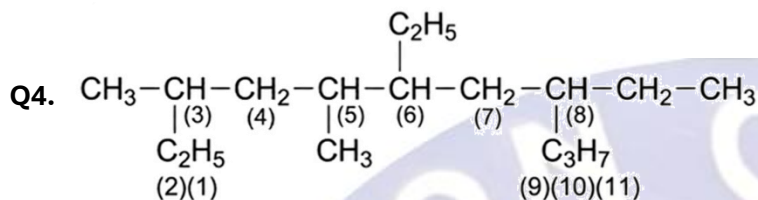
Ans I.U.P.A.C name → 2 - methyl pentane
 C.N → Iso hexane



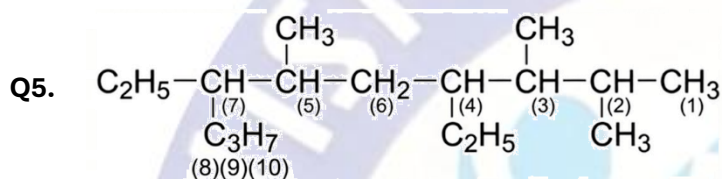
Ans 2, 5 - Dimethyl Hexane



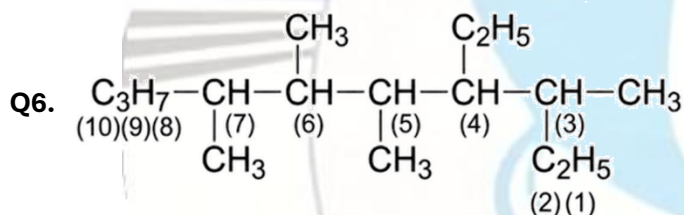
Ans I.U.P.A.C name $\rightarrow$ 2, 4, 6 – trimethyl octane



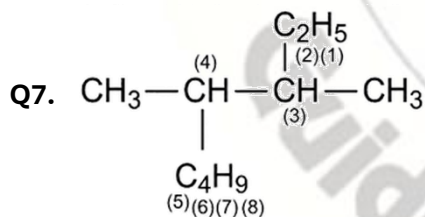
Ans I.U.P.A.C name $\rightarrow$ 6, 8 – Diethyl – 3, 5 – Dimethyl undecane



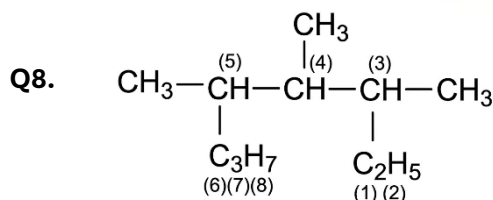
Ans I.U.P.A.C name $\rightarrow$ 4, 7 – Diethyl – 2, 3, 6 – trimethyl decane



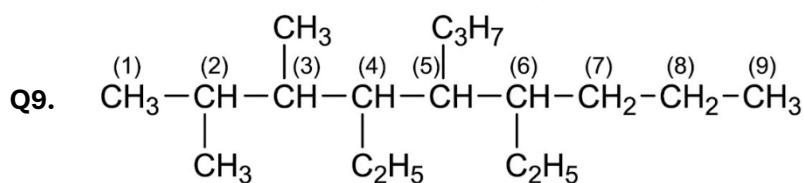
Ans I.U.P.A.C name $\rightarrow$ 4 – ethyl – 3, 5, 6, 7 – tetramethyl decane



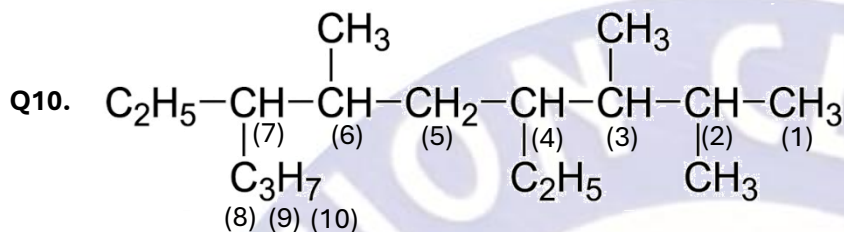
Ans I.U.P.A.C name $\rightarrow$ 3, 4 – dimethyl octane



Ans I.U.P.A.C name $\rightarrow$ 3, 4, 5 – trimethyl octane



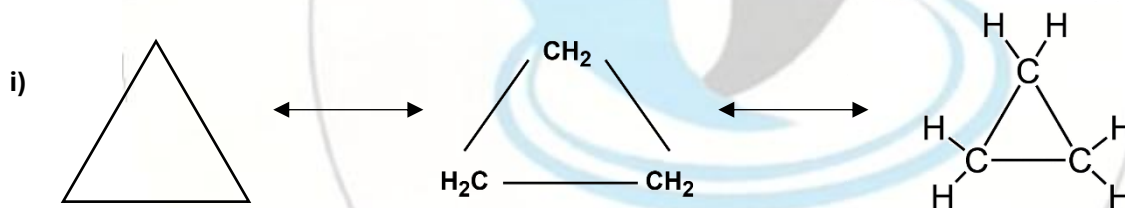
Ans I.U.P.A.C name $\rightarrow$ 4, 6 - diethyl - 2, 3 - dimethyl - 5 - propyl nonane



Ans I.U.P.A.C name $\rightarrow$ 4, 7 - Diethyl - 2, 3, 6 - trimethyl decane

- **Cyclo alkane**
 - $\downarrow$ Single Bond
 - Cyclic**
 - $\rightarrow$ Ring
 - $\rightarrow$ Minimum C - atom for a ring = 3

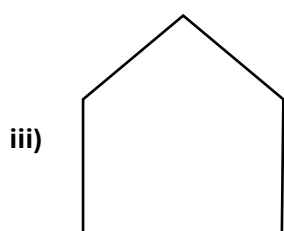
For example:-



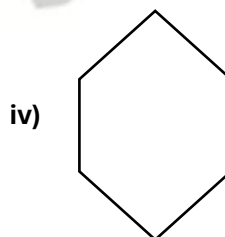
I.U.P.A.C name $\rightarrow$ Cyclo propane



I.U.P.A.C name $\rightarrow$ Cyclo butane

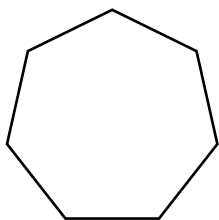


I.U.P.A.C name $\rightarrow$ Cyclo pentane



I.U.P.A.C name $\rightarrow$ Cyclo Hexane

v)

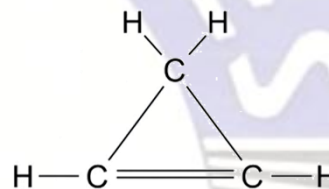
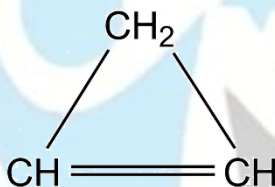


I.U.P.A.C name → Cyclo Heptane

- **Cyclo Alkene**
 - Double Bond
- **Cyclic**
 - Ring
 - Minimum C – atom for a ring = 3

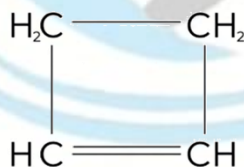
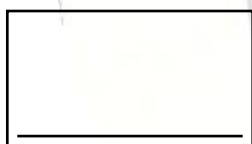
For example:-

i)



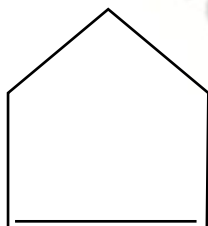
I.U.P.A.C name → Cyclo propene

ii)



I.U.P.A.C name → Cyclo butene

iii)



I.U.P.A.C name → Cyclo pentene

iv)



I.U.P.A.C name → Cyclo hexene

- **Systematic Representation**

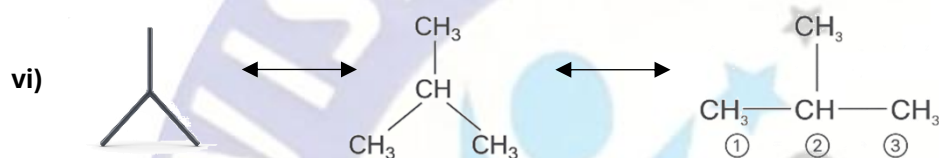
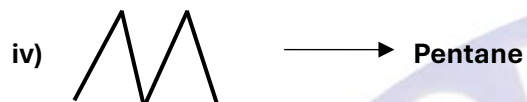
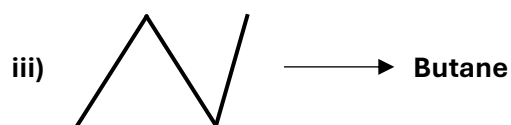
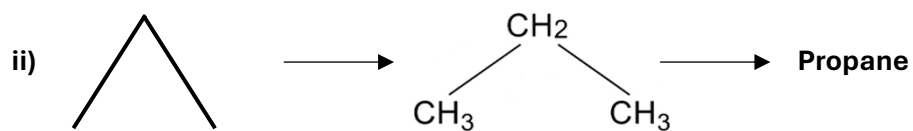
➤ In systematic representation, we represent hydrocarbon chain in line.

For example:-

i)

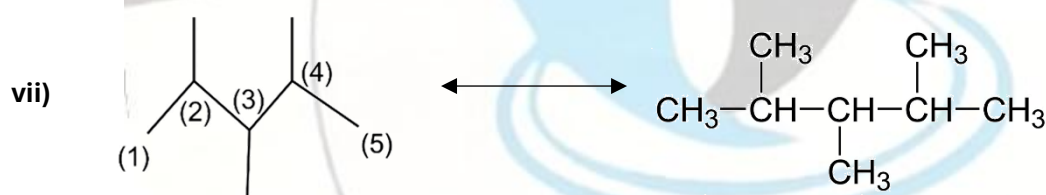


→ Ethane



C.N $\rightarrow$ Iso butane

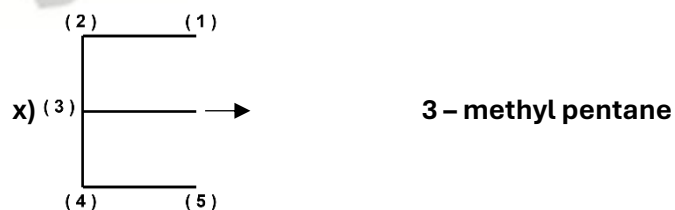
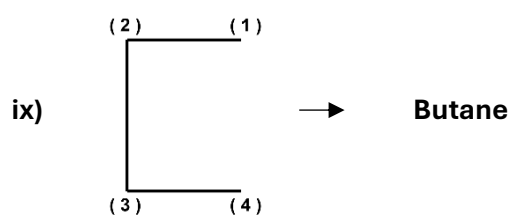
I.U.P.A.C name $\rightarrow$ 2 – methyl propane

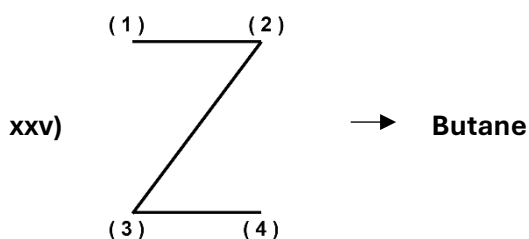
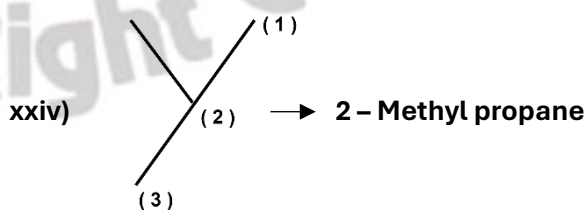
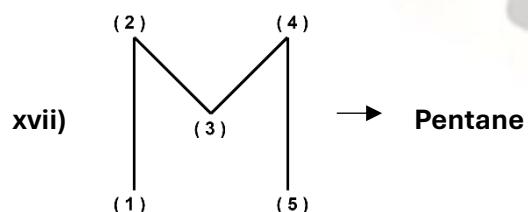
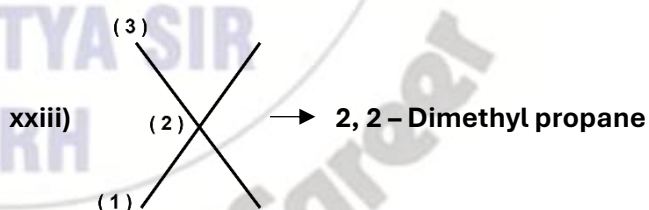
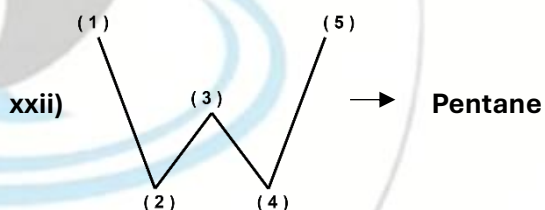
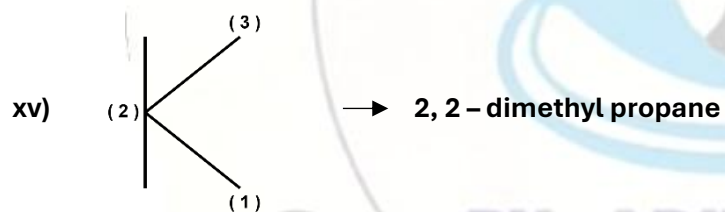
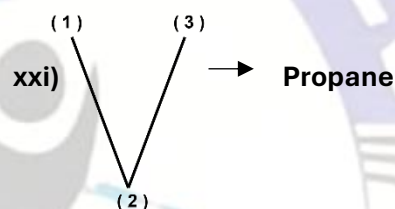
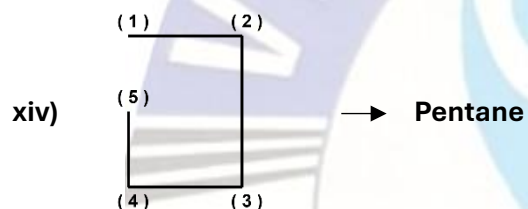
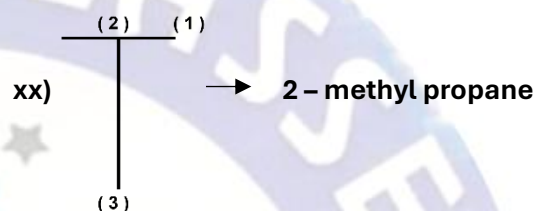
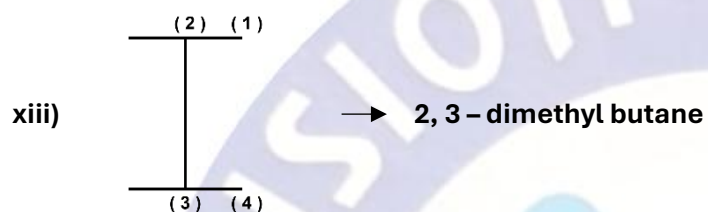
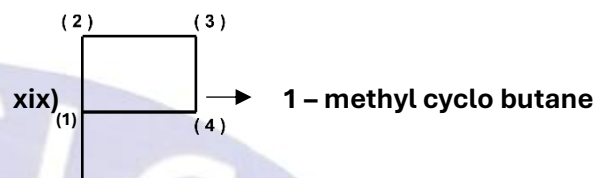
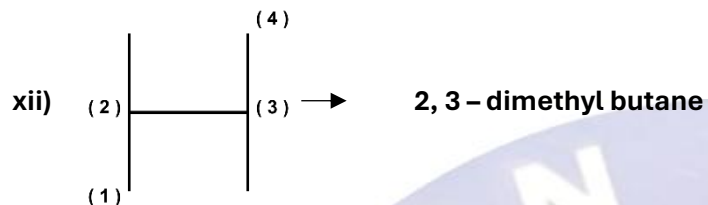
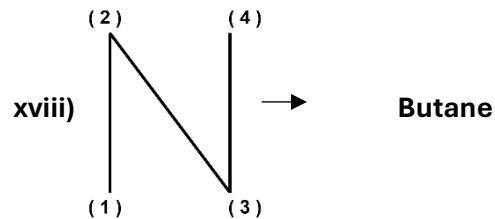
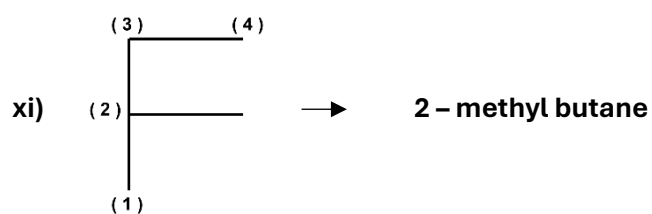


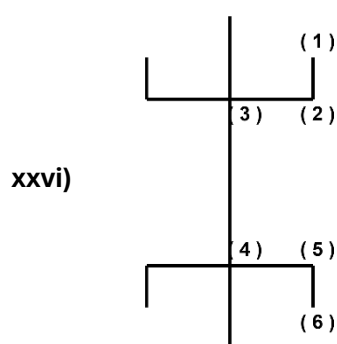
I.U.P.A.C name $\rightarrow$ 2, 3, 4 – trimethyl pentane



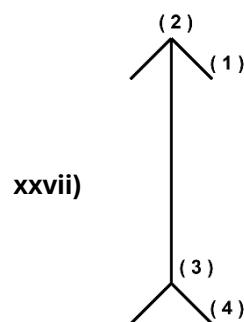
I.U.P.A.C name $\rightarrow$ 1, 2 – dimethyl cyclopropane



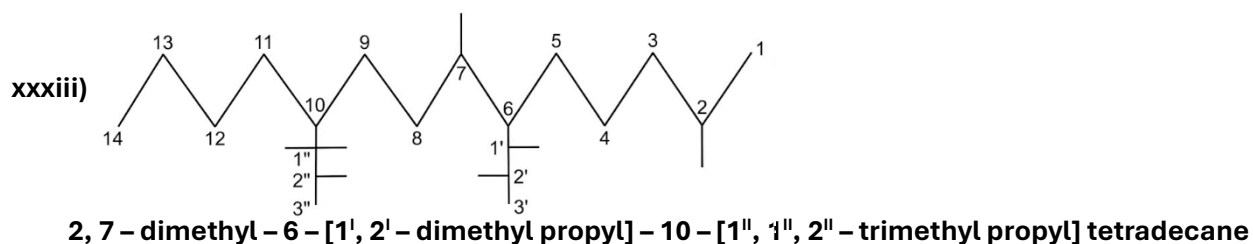
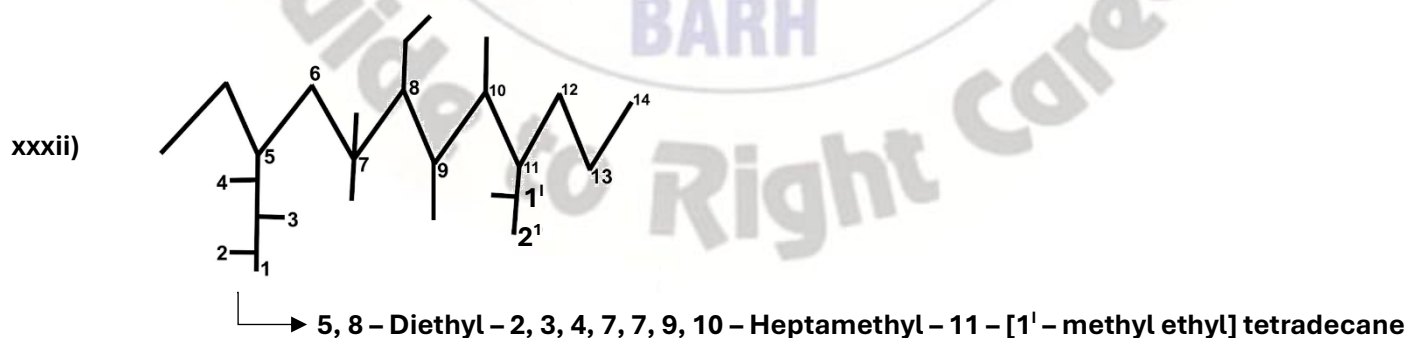
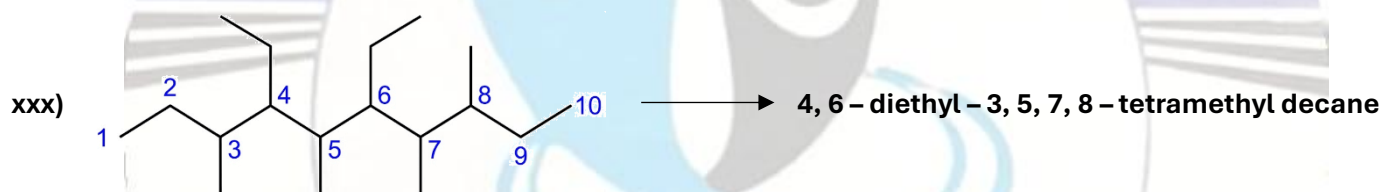
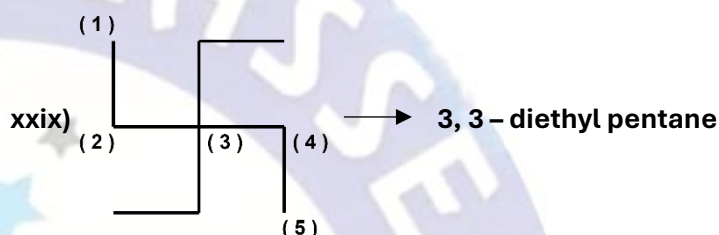
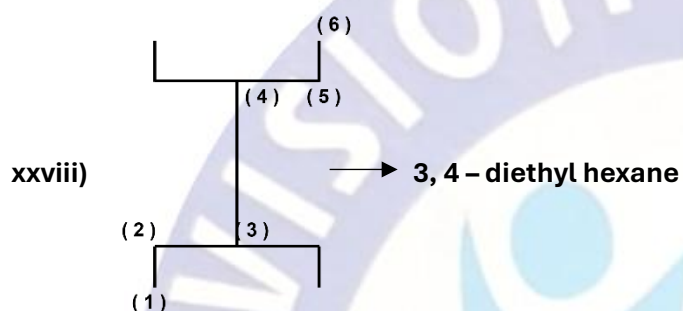




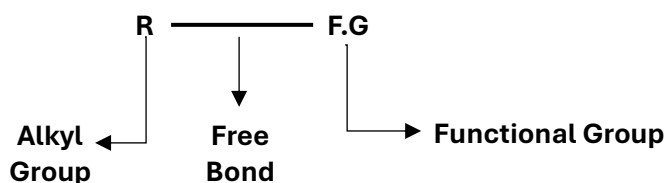
3, 4 – Diethyl – 3, 4 – dimethyl hexane



2, 3 – dimethyl butane



• **Derivative of Hydrocarbon**



• **Functional Group**

- An atom or group of atoms within a molecule that has similar chemical properties whenever it appears in various compound, called functional group.

For example:-

- ✓ Alcohol → OH
- ✓ Aldehyde → CHO
- ✓ Carboxylic acid → COOH etc.

Functional Group	Molecular Formula	Suffix
Alkene	= { Unsaturated	ene
Alkyne	≡ { Hydrocarbon	yne
Alcohol	- OH	ol
Aldehyde	- CHO	al
Ketone	- CO -	one
Carboxylic acid	- COOH	oic
Halogen	(X = F, Cl, Br, I)	Halide
Amide	- CONH ₂	amide
Ether	- O -	oxy
Ester	- COO -	oate

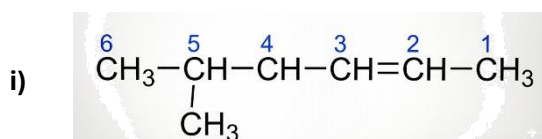
• **ALKENE** → (C = C), G.F → C_nH_{2n}

I.U.P.A.C Suffix → ene, Common suffix → ylene

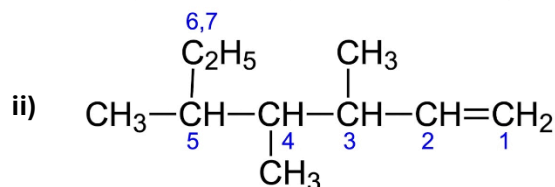
No. of C - atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 1	C ₁ H _{2×1} = CH ₂ Not possible			
n = 2	C ₂ H _{2×2} = C ₂ H ₄		Eth + ene Ethene	Eth + ylene Ethylene
n = 3	C ₃ H _{2×3} = C ₃ H ₆		Prop + ene Propene	Prop + ylene Propylene / 1 - propylene

n = 4	C_4H_8	$ \begin{array}{ccccccc} & H & & H & & H & \\ & & & & & & \\ H & -C & - & C & - & C & =C-H \\ & & & & & & \\ & H & & H & & & \end{array} $	But + ene Butene OR But - 1 - ene Or, 1 - Butene.	1 - butylene
n = 5	C_5H_{10}	$ \begin{array}{ccccccc} & & & H & & H & & H \\ & & & & & & & \\ H & & & C & = & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & & & \\ & & & H & & H & & H & & H & & H \end{array} $ OR $ \begin{array}{ccccccc} & H & & H & & & & H & & H \\ & & & & & & & & & \\ H & -C & - & C & = & C & - & C & - & C & - & H \\ & & & & & & & & & \\ & H & & & & & & H & & H \end{array} $	Pentene / Pent - 1 - ene / 1 - Pentene 2 - Pentene / Pent - 2 - ene	1 - pentylene pent - 2 - ylene
n = 6	C_6H_{12}	$ \begin{array}{ccccccc} & & & H & & H & & H & & H \\ & & & & & & & & & \\ H & & & C & = & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & & & \\ & & & H & & H & & H & & H & & H \end{array} $ OR $ \begin{array}{ccccccc} & H & & & & H & & H & & H \\ & & & & & & & & & \\ H & -C & - & C & = & C & - & C & - & C & - & C & - & H \\ & & & & & & & & & \\ & H & & H & & H & & H & & H \end{array} $ OR $ \begin{array}{ccccccc} & H & & H & & & & H & & H \\ & & & & & & & & & \\ H & -C & - & C & - & C & = & C & - & C & - & C & - & H \\ & & & & & & & & & \\ & H & & H & & H & & H & & H \end{array} $	Hexene / 1 - Hexene / Hex - 1 - ene 2 - Hexene / Hex - 2 - ene 3 - Hexene / Hex - 3 - ene	1 - Hexylene Hex - 2 - ylene Hex - 3 - ylene

Special Nomenclature



I.U.P.A.C → 5 - methyl hex - 2 - ene



I.U.P.A.C → 3, 4, 5 – trimethyl hept – 1 – ene

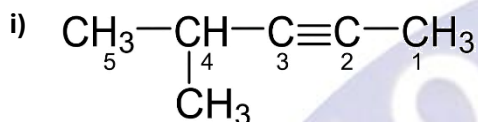
- **ALKYNE** → ($C \equiv C$), G.F = C_nH_{2n-2}

I.U.P.A.C suffix → yne, Common suffix → Acetylene

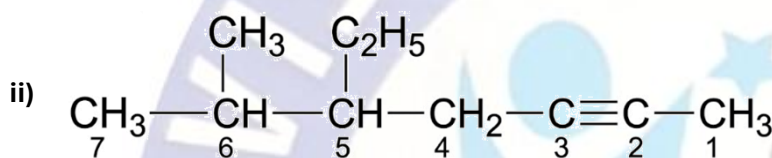
No. of C – atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 1	$C_1H_{2 \times 1 - 2} = CH_0$ Not possible			
n = 2	$C_2H_{2 \times 2 - 2} = C_2H_2$		Eth + yne Ethyne	Acetylene
n = 3	C_3H_4		Propyne / Prop – 1 – yne / 1 – Propyne	Methyl Acetylene
n = 4	C_4H_6		Butyne / But – 1 – yne / 1 – Butyne 2 – Butyne / But – 2 – yne	Ehtyl Acetylene Dimethyl Acetylene
n = 5	C_5H_8		Pentyne / 1 – Pentyne / Pent – 1 – yne	Propyl Acetylene

		OR $\begin{array}{c} \text{H} & & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}-\text{C}\equiv\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \\ \text{H} & & \text{H} & \text{H} \end{array}$	Pent – 2 – yne / 2 – Pentyne	Ehtyl methyl Acetylene
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Special Nomenclature



I.U.P.A.C → 4 – methyl – 2 – Pentyne



I.U.P.A.C → 5 – Ethyl – 6 – methyl – 2 – heptyne

- **ALCOHOL** [R – OH] → $\text{C}_n\text{H}_{2n+1} + \text{OH}$

I.U.P.A.C name → Alkane + ol → Alkanol, Common name → Alkyl + Alcohol

No. of C – atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 1	CH_3OH	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$	Methane + ol Methanol	Methyl alcohol
n = 2	$\text{C}_2\text{H}_5\text{OH}$	$\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C} & -\text{C}-\text{O}-\text{H} \\ & \\ \text{H} & \text{H} \end{array}$	Ethanol	Ethyl alcohol
n = 3	$\text{C}_3\text{H}_7\text{OH}$	$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C} & -\text{C}-\text{C}-\text{O}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$	Propanol / Propane – 1 – ol / 1 – propanol	Propyl alcohol

		OR $ \begin{array}{ccccc} & \text{H} & \text{H} & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & \\ & \text{H} & \text{OH} & \text{H} & \end{array} $ (Carbons labeled 3, 2, 1 from left to right)	2 – Propanol / Propane – 2 – ol	Propyl alcohol
n = 4	C ₄ H ₉ OH	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{OH} & \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $	Butanol	Butyl Alcohol
n = 5	C ₅ H ₁₁ OH	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{OH} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \end{array} $	Pentanol	Pentyl alcohol

• **HALOGEN [R – X, X = (F, Cl, Br, I)]**

I.U.P.A.C name → Halo alkane
 Common name → Alkyl Halide

No. of C – atom	M.F	S.F (Structural Formula)	I.U.P.A.C name	Common Name
n = 1	CH ₃ F	$ \begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{F} \\ \\ \text{H} \end{array} $	Fluoro methane	Methyl Fluoride
n = 2	C ₂ H ₅ F	$ \begin{array}{ccccc} & \text{H} & & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{F} \\ & & & & \\ & \text{H} & & \text{H} & \end{array} $	Fluoro ethane	Ethyl Fluoride
n = 3	C ₃ H ₇ Cl	$ \begin{array}{ccccc} & \text{H} & & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} - \text{Cl} \\ & & & & \\ & \text{H} & & \text{H} & \text{H} \end{array} $	Chloro Propane	Propyl Chloride

n = 4	C_4H_9Br	$ \begin{array}{cccc} H & H & H & H \\ & & & \\ H-C & -C & -C & -C-Br \\ & & & \\ H & H & H & H \end{array} $	Bromo Butane	Butyl Bromide
n = 5	$C_5H_{11}I$	$ \begin{array}{ccccc} H & H & H & H & H \\ & & & & \\ H-C & -C & -C & -C & -C-I \\ & & & & \\ H & H & H & H & H \end{array} $	Iodo Pentane	Pentyl Iodide
n = 6	$C_6H_{13}Cl$	$ \begin{array}{cccccc} H & H & H & H & H & H \\ & & & & & \\ H-C & -C & -C & -C & -C & -C-Cl \\ & & & & & \\ H & H & H & H & H & H \end{array} $	Chloro Hexane	Hexyl Chloride

• **ETHER (R - O - R')**

I.U.P.A.C name → Prefix + Oxy Alkane

Common name → Alkyl Alkyl ether

Note:

Less C - atoms में Oxy

More C - atoms में ane

No. of C - atom	M.F	S.F	I.U.P.A.C name	Common Name
1 R = CH_3 R' = CH_3	CH_3OCH_3 Or $(CH_3)_2O$	$ \begin{array}{ccc} H & & H \\ & & \\ H-C & -O- & C-H \\ & & \\ H & & H \end{array} $	Methoxy Methane	Dimethyl ether
2 R = CH_3 R' = C_2H_5	$CH_3OC_2H_5$	$ \begin{array}{ccc} H & & H & H \\ & & & \\ H-C & -O- & C & -C-H \\ & & & \\ H & & H & H \end{array} $	Methoxy Ethane	Ethyl Methyl ether
3 R = C_2H_5 R' = C_2H_5	$C_2H_5OC_2H_5$	$ \begin{array}{ccc} H & H & & H & H \\ & & & & \\ H-C & -C & -O- & C & -C-H \\ & & & & \\ H & H & & H & H \end{array} $	Ethoxy Ethane	Diethyl ether

4 R = C ₂ H ₅ R' = C ₃ H ₇	C ₂ H ₅ OC ₃ H ₇	$ \begin{array}{ccccccc} & \text{H} & \text{H} & & \text{H} & \text{H} & \text{H} \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{O} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & \\ & \text{H} & \text{H} & & \text{H} & \text{H} & \text{H} \end{array} $	Ethoxy Propane	Ethyl propyl ether
5 R = C ₃ H ₇ R' = C ₄ H ₉	C ₃ H ₇ OC ₄ H ₉	$ \begin{array}{cccccccc} & \text{H} & \text{H} & \text{H} & & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{O} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & & & \\ & \text{H} & \text{H} & \text{H} & & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	Propoxy Butane	Butyl Propyl ether
6 R = C ₃ H ₇ R' = C ₃ H ₇	C ₃ H ₇ OC ₃ H ₇	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & & \text{H} & \text{H} & \text{H} \\ & & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{O} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & & \\ & \text{H} & \text{H} & \text{H} & & \text{H} & \text{H} & \text{H} \end{array} $	Propoxy Propane	Dipropyl ether

- The carbon containing functional group uses special prefix for common name

No. of C – atom	Special Prefix (Common name)
1	Form
2	Acet
3	Propion
4	Butyr
5	Valer ↗ Iso ↘ Neo
6	Capro

- ALDEHYDE** → R – CHO, Suffix – al

I.U.P.A.C name → Alkane + al → Alkanal

Common name → Special Prefix + aldehyde

No. of C – atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 0	HCHO	$ \begin{array}{c} \text{O} \\ \\ \text{H} - \text{C} - \text{H} \end{array} $	Methanal	Form + aldehyde Formaldehyde
n = 1	CH ₃ CHO	$ \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{H} \\ \\ \text{H} \end{array} $	Ethanal	Acet + aldehyde Acetaldehyde

n = 2	C_2H_5CHO	$ \begin{array}{c} H & H & O \\ & & \\ H-C & -C & -C-H \\ & & \\ H & H & \end{array} $	Propanal	Propion + aldehyde Propionaldehyde
n = 3	C_3H_7CHO	$ \begin{array}{c} H & H & H & O \\ & & & \\ H-C & -C & -C & -C-H \\ & & & \\ H & H & H & \end{array} $	Butanal	Butyraldehyde
n = 4	C_4H_9CHO	$ \begin{array}{c} H & H & H & H & O \\ & & & & \\ H-C & -C & -C & -C & -C-H \\ & & & & \\ H & H & H & H & \end{array} $	Pentanal	Valeraldehyde
n = 5	$C_5H_{11}CHO$	$ \begin{array}{c} H & H & H & H & H & O \\ & & & & & \\ H-C & -C & -C & -C & -C & -C-H \\ & & & & & \\ H & H & H & H & H & \end{array} $	Hexanal	Ca proaldehyde

• **CARBOXYLIC ACID (R - COOH)**

I.U.P.A.C name → Alkane + oic acid → Alkanoic acid

Common name → (Special prefix + ic) acid

No. of C - atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 0	$HCOOH$	$ \begin{array}{c} O \\ \\ H-C-O-H \end{array} $	Methanoic acid	Formic acid
n = 1	CH_3COOH	$ \begin{array}{c} H & O \\ & \\ H-C & -C \\ & \backslash \\ H & O-H \end{array} $	Ethanoic acid	Acetic acid

n = 2	C ₂ H ₅ COOH	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} $	Propanoic acid	Propionic acid
n = 3	C ₃ H ₇ COOH	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $	Butanoic acid	Butyric acid
n = 4	C ₄ H ₉ COOH	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Pentanoic acid	Valeric acid
n = 5	C ₅ H ₁₁ COOH	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Hexanoic acid	Caproic acid

• **AMIDE** → [R – CONH₂]

I.U.P.A.C name → Alkane + amide → Alkanamide

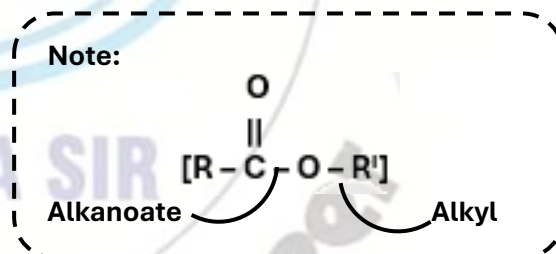
Common name → Special prefix + amide

No. of C – atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 0	HCONH ₂	$ \begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{N} \begin{array}{l} \nearrow \text{H} \\ \searrow \text{H} \end{array} \end{array} $	Methane + amide Methanamide	Formamide Form + amide
n = 1	CH ₃ CONH ₂	$ \begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{N} \begin{array}{l} \nearrow \text{H} \\ \searrow \text{H} \end{array} \\ \\ \text{H} \end{array} $	Ethanamide	Acetamide

n = 2	$C_2H_5CONH_2$	$ \begin{array}{c} H & H & O \\ & & \\ H-C & -C & -C-N \\ & & / \quad \backslash \\ H & H & H & H \end{array} $	Propanamide	Propionamide
n = 3	$C_3H_7CONH_2$	$ \begin{array}{c} H & H & H & O \\ & & & \\ H-C & -C & -C & -C-N \\ & & & / \quad \backslash \\ H & H & H & H & H \end{array} $	Butanamide	Butyramide
n = 4	$C_4H_9CONH_2$	$ \begin{array}{c} H & H & H & H & O \\ & & & & \\ H-C & -C & -C & -C & -C-N \\ & & & & / \quad \backslash \\ H & H & H & H & H & H \end{array} $	Pentanamide	Valeramide
n = 5	$C_5H_{11}CONH_2$	$ \begin{array}{c} H & H & H & H & H & O \\ & & & & & \\ H-C & -C & -C & -C & -C & -C-N \\ & & & & & / \quad \backslash \\ H & H & H & H & H & H & H \end{array} $	Hexanamide	Caproamide

- **ESTER** $[R-\overset{\overset{O}{||}}{C}-O-R']$

I.U.P.A.C name → Alkyl alkane + oate → Alkyl Alkanoate
 Common name → Alkyl Special Prefix + ate



No. of C-atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 0, n = 1 Or, R = H, R' = CH ₃	HCOOCH ₃	$ \begin{array}{c} O & H \\ & \\ H-C & -O-C-H \\ & \\ & H \end{array} $	Methyl Methanoate	Methyl Formate
R = CH ₃ R' = CH ₃	CH ₃ COOCH ₃	$ \begin{array}{c} H & O & H \\ & & \\ H-C & -C & -O-C-H \\ & & \\ H & & H \end{array} $	Methyl Ethanoate	Methyl Acetate

$R = \text{CH}_3$ $R' = \text{C}_2\text{H}_5$	$\text{CH}_3\text{COOC}_2\text{H}_5$		Ethyl Ethanoate	Ethyl Acetate
$R = \text{C}_2\text{H}_5$ $R' = \text{C}_2\text{H}_5$	$\text{C}_2\text{H}_5\text{COOC}_2\text{H}_5$		Ethyl Propanoate	Ethyl Propionate
$R = \text{C}_2\text{H}_5$ $R' = \text{C}_3\text{H}_7$	$\text{C}_2\text{H}_5\text{COOC}_3\text{H}_7$		Propyl Propanoate	Propyl Propionate

- **KETONE** [$\text{R}-\text{C}-\text{R}'$], Don't use specific prefix

I.U.P.A.C name $\rightarrow$ Alkane + one $\rightarrow$ Alkanone

Common name $\rightarrow$ Alkyl Alkyl ketone

No. of C-atom	M.F	S.F	I.U.P.A.C name	Common Name
$R = \text{CH}_3$ $R' = \text{CH}_3$	CH_3COCH_3		Propanone or, Propane - 2 - one or, 2 - propanone	Dimethyl Ketone or, Acetone
$R = \text{CH}_3$ $R' = \text{C}_2\text{H}_5$	$\text{CH}_3\text{COC}_2\text{H}_5$		2 - Butanone	Ethyl methyl ketone
$R = \text{CH}_3$ $R' = \text{C}_3\text{H}_7$	$\text{CH}_3\text{COC}_3\text{H}_7$		Pentanone 2 - pentanone	Methyl propyl ketone
$R = \text{C}_2\text{H}_5$ $R' = \text{C}_3\text{H}_7$	$\text{C}_2\text{H}_5\text{COC}_3\text{H}_7$		3 - Hexanone	Ethyl Propyl ketone

$R = \text{CH}_3$ $R' = \text{C}_4\text{H}_9$	$\text{CH}_3\text{COC}_4\text{H}_9$	$ \begin{array}{ccccccc} & \text{H} & & \text{O} & & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & & \\ & \text{H} & & & & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	2 – Hexanone	Butyl methyl ketone
$R = \text{CH}_3$ $R' = \text{C}_5\text{H}_{11}$	$\text{CH}_3\text{COC}_5\text{H}_{11}$	$ \begin{array}{ccccccc} & \text{H} & & \text{O} & & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & & & \\ & \text{H} & & & & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	2 – Heptanone	methyl pentyl ketone

• **AMINE** → $\text{R} - \text{NH}_2$

I.U.P.A.C name → Alkane + amine → Alkanamine

Common name → Alkyl amine

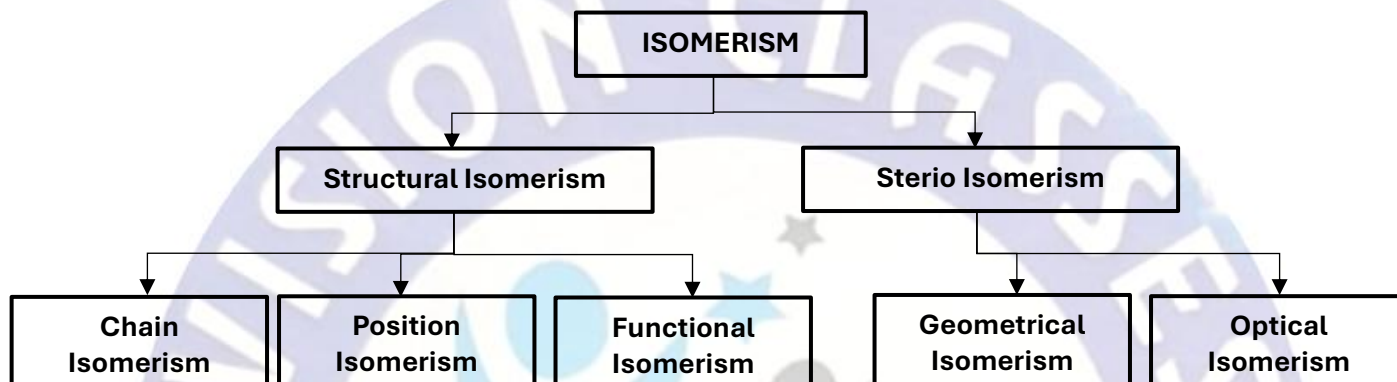
No. of C – atom	M.F	S.F	I.U.P.A.C name	Common Name
n = 1	CH_3NH_2	$ \begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{N} \\ \quad \diagup \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $	Methanamine	Methyl amine
n = 2	$\text{C}_2\text{H}_5\text{NH}_2$	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{N} \\ \quad \quad \diagup \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Ethanamine	Ethyl amine
n = 3	$\text{C}_3\text{H}_7\text{NH}_2$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{N} \\ \quad \quad \quad \diagup \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Propanamine	Propyl amine
n = 4	$\text{C}_4\text{H}_9\text{NH}_2$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{N} \\ \quad \quad \quad \quad \diagup \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Butanamine	Butyl amine
n = 5	$\text{C}_5\text{H}_{11}\text{NH}_2$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{N} \\ \quad \quad \quad \quad \quad \diagup \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Pentanamine	Pentyl amine
n = 6	$\text{C}_6\text{H}_{13}\text{NH}_2$	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{N} \\ \quad \quad \quad \quad \quad \quad \diagup \quad \diagdown \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $	Hexanamine	Hexyl amine

ISOMERISM

Isos + Merus
↓ ↓
Some Part

- **Definition**

- When two or more carbon compounds have same molecular formula but different physical or chemical properties are called Isomers and this phenomenon is known as Isomerism.



1. Structural Isomerism

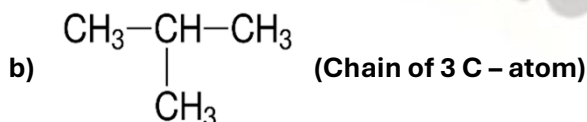
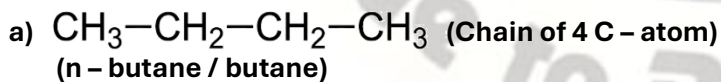
- Structural Isomerism are those Isomers in which the atom are completely arranged in a different order with the same molecular formula i.e, atom have different arrangement, this phenomenon is known as structural Isomerism.
- $A-B-C-D / A-C-B-D$

- **Chain Isomerism**

- The Isomerism arises due to the change in arrangement of carbon atom in the chain, called chain isomerism.
- Chain isomerism are also called 'Nuclear Isomerism'.

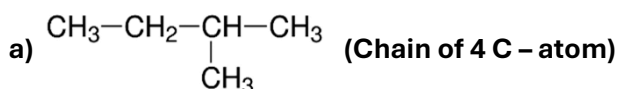
For example:-

- ✓ C_4H_{10} (Butane) has 2 Isomers

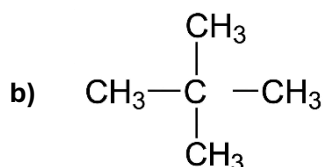


2 – methyl propane

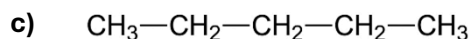
- ✓ C_5H_{12} (Pentane) has 3 Isomers



2 – methyl butane



2, 2 – dimethyl propane



(n – Pentane / Pentane)

▪ **Position Isomerism**

- The Isomers arises due to the different position in functional group, called position isomerism.

In Position Isomerism

- a) Same molecular formula.
- b) Same length of carbon chain.
- c) Same functional group but different position.

For example:-

$\text{C}_3\text{H}_7\text{OH}$ has two isomers.

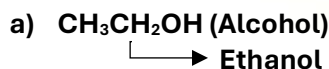


▪ **Functional Isomerism**

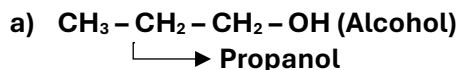
- When two or more organic compounds having same molecular formula but different functional group are called functional isomers and this phenomenon is known as functional isomerism.

For example:-

✓ $\text{C}_2\text{H}_5\text{OH}$

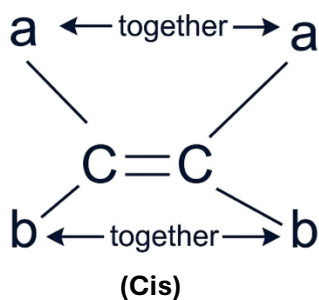


✓ $\text{C}_3\text{H}_7\text{OH}$

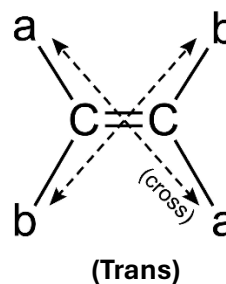


2. **Stereo Isomerism**

- Isomers having same molecular formula, same sequence in bonding of there atom i.e, same structure but different special arrangement called Sterio Isomerism.

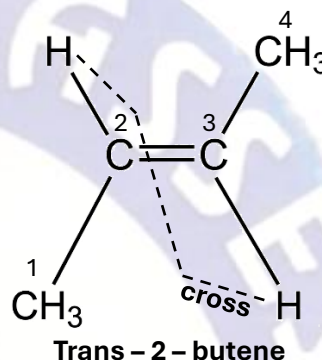
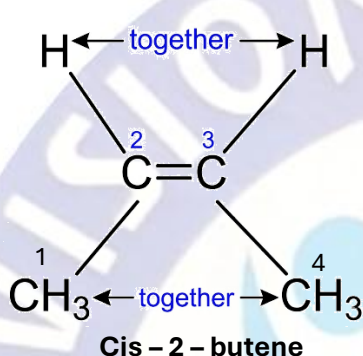


OR



▪ **Geometrical Isomerism**

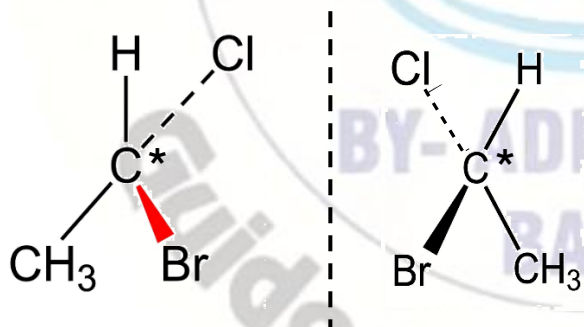
- That isomerism in which the two different – different groups are added with each carbon atom which are connected by double bond.



▪ **Optical Isomerism**

- That isomerism in which four different atoms or, compound attached with a carbon atom, these are called optical isomers and the phenomenon is known as optical isomerism.
- Optical isomers are mirror image to each other.

For example:-



Note:

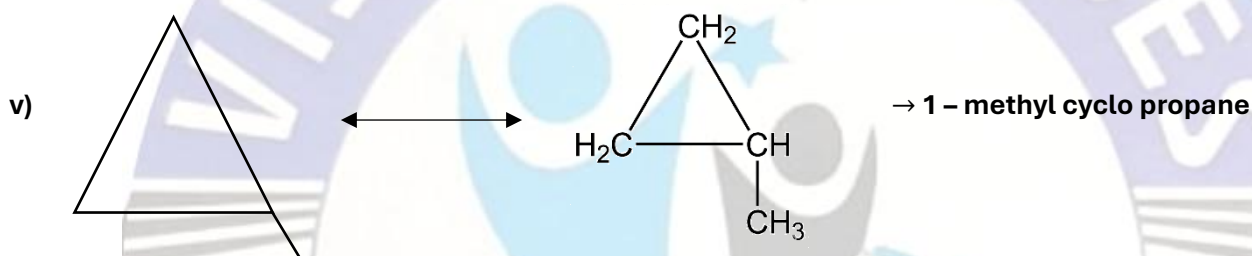
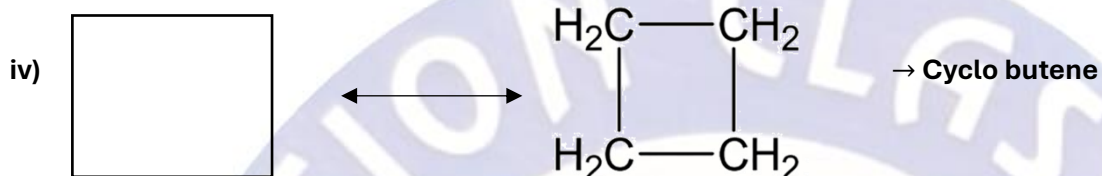
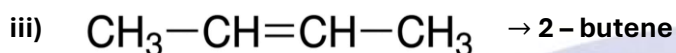
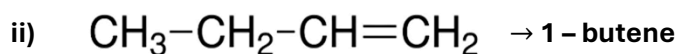
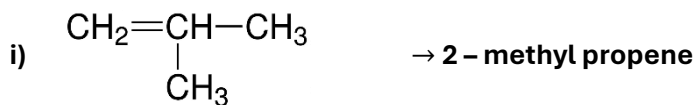
i) → above the plane
 Wedge

ii) (-----) → below the board
 Dotted line

iii) C* → Asymmetric carbon atom
 When four different groups / atoms attached with a carbon.

iv) → Bold Bond [On the plane]

Q. How many possible Isomers of C_4H_8 .

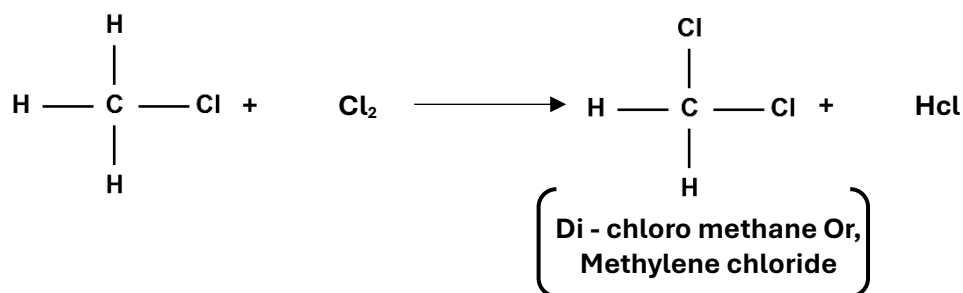
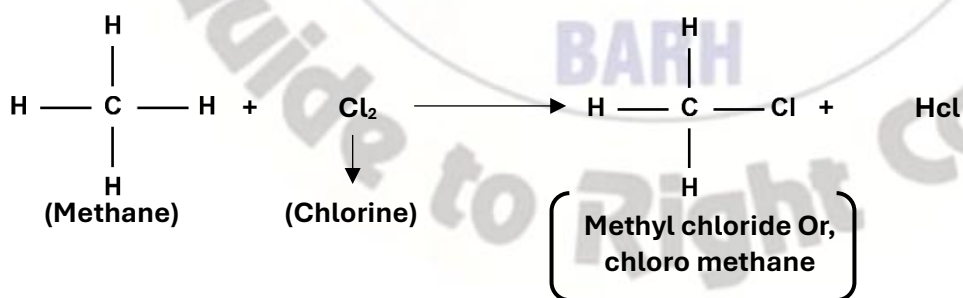


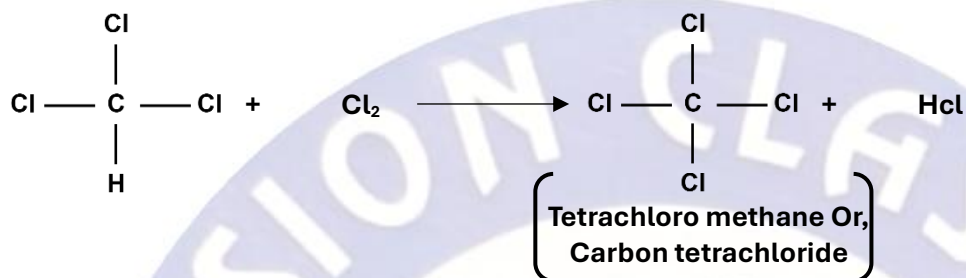
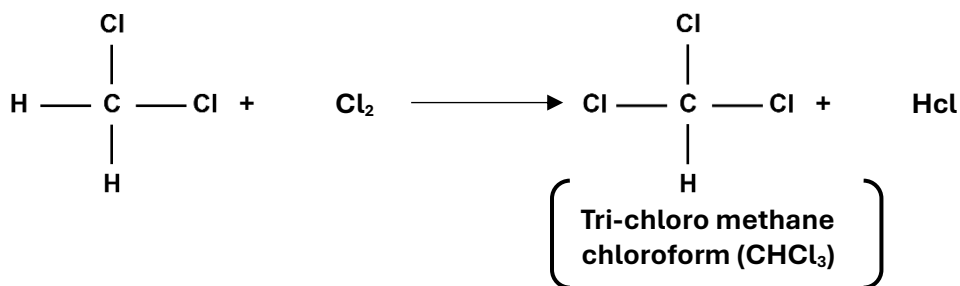
• **Substitution Reaction**

- That reaction in which an atom or group of atoms in a molecule is replaced or, substituted by different atom or, group of atom is called substitution reaction.
- A substitution reaction can also be called a single displacement reaction.

For example:-

- ✓ Substitution of hydrogen by chlorine in alkane in the presence of sunlight.





The above reaction is also called chlorination reaction.

✓ Substitution reaction is also called Halogenation.

• **Addition Reaction**

➤ Addition reaction is that reaction in which an **atom (H₂/Cl₂/O₂)** is added to unsaturated hydrocarbon to obtain a saturated hydrocarbon is called addition reaction.

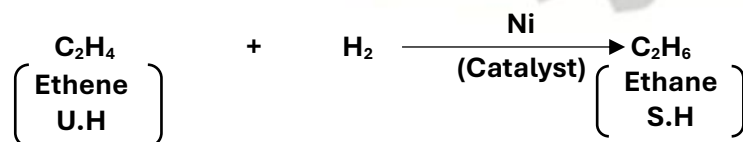
✓ It is only applicable for alkene and alkyne.



• **Hydrogenation**

➤ That addition reaction in which hydrogen is added to unsaturated hydrocarbon to obtain a saturated hydrocarbon.

For example:-



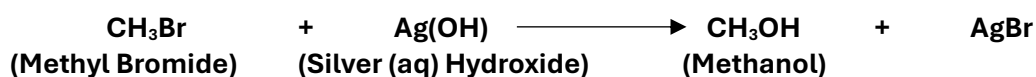
ALCOHOL

• **Formation**

➤ Alcohol is generally formed by hydrolysis of alkyl bromide by the aqueous humid silver hydroxide.

For example:-

✓ Formation of methanol



▪ Properties of Methanol

Case 1: Physical properties

- i) It is a colourless liquid.
- ii) It is poison.
- iii) It's boiling point is very low (64.6°C).
- iv) It is totally soluble in water.

Case 2: Chemical properties

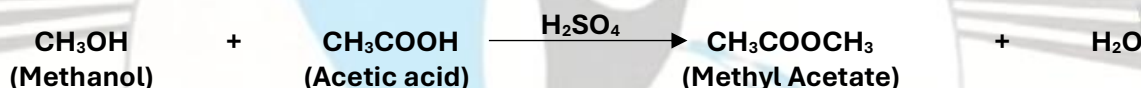
i) Reaction with sodium:

- It reacts with sodium to form sodium methoxide and evaluate hydrogen gas.



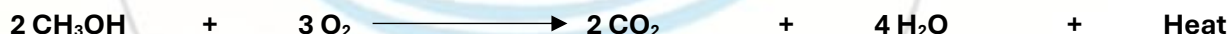
ii) Reaction with Acetic acid / Ethanoic Acid:

- Methanol reacts with Ethanoic acid in the presence of concentrated H_2SO_4 to form methyl acetate (ester).

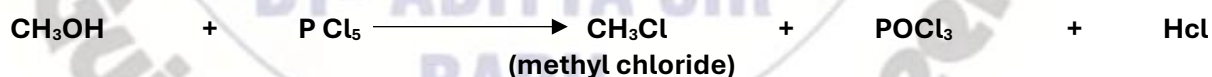


iii) Combustion:

- Methanol is a combustible substance. It burns with blue flame and produce CO_2 & H_2O .



iv) Reaction with phosphorus pentachloride



Methanol reacts with phosphorus pentachloride to form methyl chloride.

▪ Uses of methanol

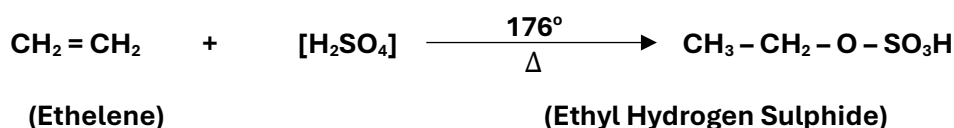
- i) **Methanol is used in making paints.**
- ii) It is used as solvent in **barnish** and polish.
- iii) It is used in making dyes, medicine etc.

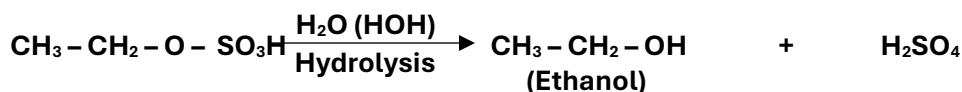
ETHANOL / ETHYL ALCOHOL



• Formation

- Ethyl alcohol is prepared by reaction of ethelene gas and concentrated H_2SO_4 .





▪ **Properties of ethanol**

Case 1: Physical properties

- i) It is a colourless liquid having pleasant smell.
- ii) It has burning taste and volatile.
- iii) It has low boiling point (78°C), average 70°C.
- iv) It is miscible with water in all proportion.
- v) It is neutral compound.
- vi) It is lighter than water.
- vii) It is non – conductor of electricity because it does not contain ions.

Case 2: Chemical properties

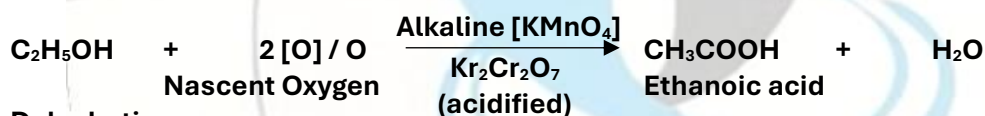
i) **Combustion:**

- Ethanol is highly inflammable. It burns with blue flame rapidly to form carbon – dioxide water vapour (volatile), heat and light.



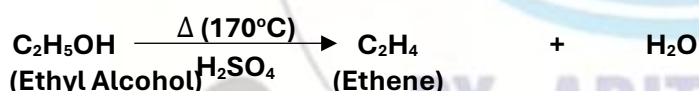
ii) **Oxidation:**

- Ethanol is oxidized and heating with alkaline potassium permanganate solution (acidic solution) to form ethanoic acid.



iii) **Dehydration:**

- On heating with excess of concentrated sulphuric acid at 170°C ethanol dehydrated to form ethene.



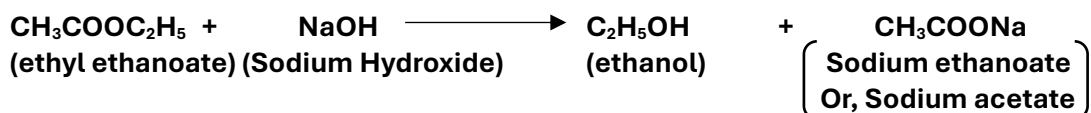
iv) **Reaction with ethanoic acid:**

- Ethanol reacts with ethanoic acid on warming in the presence of few drops of concentrated sulphuric acid to form a sweet smelling ester (ethyl ethanoate), this reaction is called esterification.



Note:

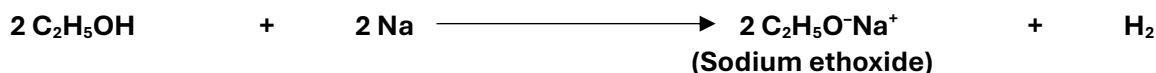
- When mixture is treated with sodium hydroxide solution, the ester is transformed back into alcohol and sodium ethanoate.



- The above reaction is an example of hydrolysis of an ester with an alkali such a reaction is called saponification.

v) **Reaction with Sodium:**

- Ethanol reacts with sodium to form sodium ethoxide.



Note:

- i) 100% pure alcohol is called ethanol or absolute alcohol.
- ii) Ethanol contains 5% of water is called rectified spirit.

▪ **Uses of Ethanol**

- i) Ethanol is partially used in beer, wine, whisky and other liquors.

Substances	Present alcohol %
Bear	3% to 6%
Wine	10% to 20%
Whisky	50%

- ii) In the formation of ethylene, chloroform, ether, acetic acid etc.
- iii) In the formation of methylated spirit which are used in making the Barnish.
- iv) In the formation of tonics, medicine, dry syrup, paint, dyes, soap synthetic rubber etc.
- v) It has lower freezing point than water so it is used as antifreezer in the radiator of the vehicle in the cold countries.
- vi) It is widely used in industries as the solvent liquors barnishes, perfume and flavorings.
- vii) Power alcohol is used as fuel in the vehicle.
- viii) In the preservation of dead body, wood etc

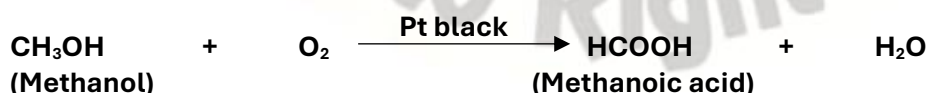
Note:

- Methanol is a poison in a small amount of drinking, causes blindness and end with death because methanol damages the optical nerve and oxidized to methanol in the liver of a person which reacts, rapidly with the components of the cells, stop functioning which leads to death of the person.

CARBOXYLIC ACID

- Carboxylic acid is formed by oxidization process of alcohol.

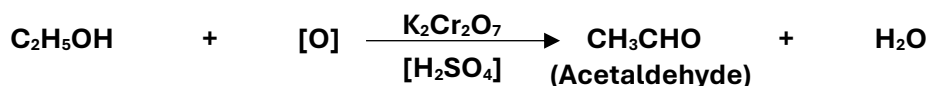
Case 1: Formation of Methanoic acid

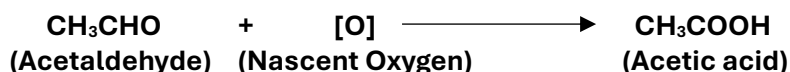


Case 2: Formation of Ethanoic Acid

▪ **Preparation / Formation**

- When ethyl alcohol is oxidised in the presence of potassium dichromate and concentrated H_2SO_4 to formed Acetic acid.





▪ **Properties of Ethanoic acid**

Case 1: Physical Properties:

- i) It is a colourless liquid.
- ii) It is sour to taste.
- iii) It's boiling point is more than water (118°C) and freezing point (289K) $\approx$ 290K.
- iv) It is miscible in water with all proportions.
- v) It freezes on cooling to colourless ice like solid so pure ethanoic acid is called glacial ethanoic acid.

Case 2: Chemical Properties:

- i) It turns blue litmus paper to red.
- ii) It is weak acid and having pH about 4.
- iii) Reaction with sodium carbonate and sodium hydrogen carbonate.

▪ **Reaction with carbonate**

➤ Ethanoic acid reacts with carbonate or, bicarbonate to form sodium ethanoate and CO₂.



▪ **Reaction with Sodium Hydroxide**

➤ Ethanoic acid reacts with sodium hydroxide to form salt and water.



▪ **Reaction with ethyl alcohol**

➤ Reaction with ethyl alcohol in the presence of concentrated sulphuric acid to form sweet and smelling ester.



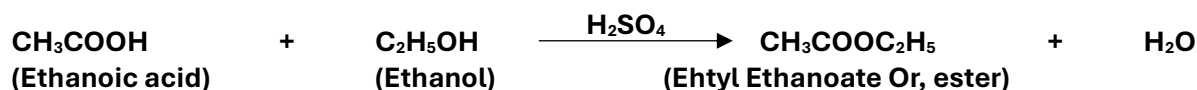
▪ **Uses of Acetic acid**

- i) In the form of mild acid.
- ii) In the form of fertilizers.
- iii) In the form of cold drinks, Medicine, perfumes, soap etc.
- iv) Widely used in textile industries.
- v) In the form of vinegar.
- vi) In the form of uses to preserve food like as pickles.
- vii) It is used in the preparation of white peris, peris green, a bright blue pigment used mainly as wood preservatives.

→ Pure acetic acid is known as glacial acetic acid having 99% acetic acid at temperature of 17°C.

▪ **Preparation of Ester (Esterification)**

- When Acetic acid reacts with ethanol in the presence of concentrated H_2SO_4 . Then ester and water is formed.



SOAP

• **Soap**

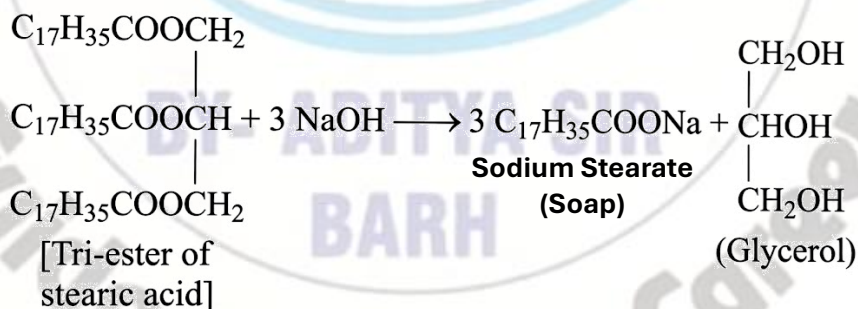
- A Soap is a sodium or potassium salt of a long chain carboxylic acid (highly fatty acid).

For example:-

Highly fatty Acid	Soap
$C_{15}H_{31}COOH$ (Palmitic acid)	$C_{15}H_{31}COONa$ (Sodium Palmitate)
$C_{17}H_{33}COOH$ (Oleic acid)	$C_{17}H_{33}COONa$ (Sodium Oleate)
$C_{17}H_{35}COOH$ (Stearic acid)	$C_{17}H_{35}COONa$ (Sodium Stearate)
$C_{17}H_{31}COOH$ (Cinoleic acid)	$C_{17}H_{31}COONa$ (Sodium Cinoleate)

▪ **Preparation of Soap**

- A soap is made from animal fats or vegetables oils. When fats and oil are heated with sodium hydroxide solution. They split into sodium salt of highly fatty acid which is called soap and other is glycerol.



• **Saponification**

- The process of splitting ester like as fats or oil using alkalis like as NaOH to form soap, called saponification.

OR

The process of forming of a soap is called saponification.



▪ **Preparation of soap in the laboratory**

- First we take 20ml of mustard oil in a beaker. An equal volume of concentrated sodium hydroxide solution is added to it. The mixture is heated slowly to boil and keep boiling for about 10 minutes. About 5g of

sodium chloride (NaCl) is added and mixture is allowed to cool. A crust is formed on the surface of liquid. This crust is known as soap.

Note:

- Mustered Oil contain tri – ester of Oleic acid so the soap is i.e, obtained is sodium oleate.

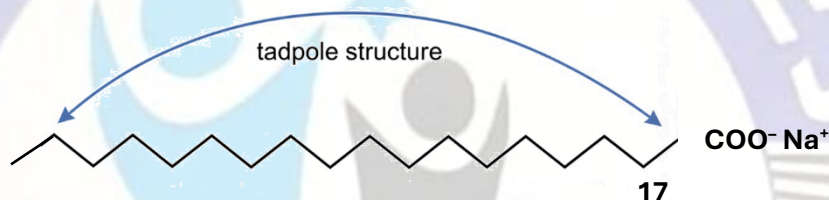
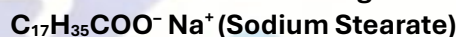
Q. Why do we added common salt (NaCl) in the formation of soap ?

Ans Common salt is added to the mixture to make the soap come out from the mixture. Because common salt decreases the solubility of soap in the solution due to which all precipitate of soap come out on the surface of solution.

• **Structure of Soap**

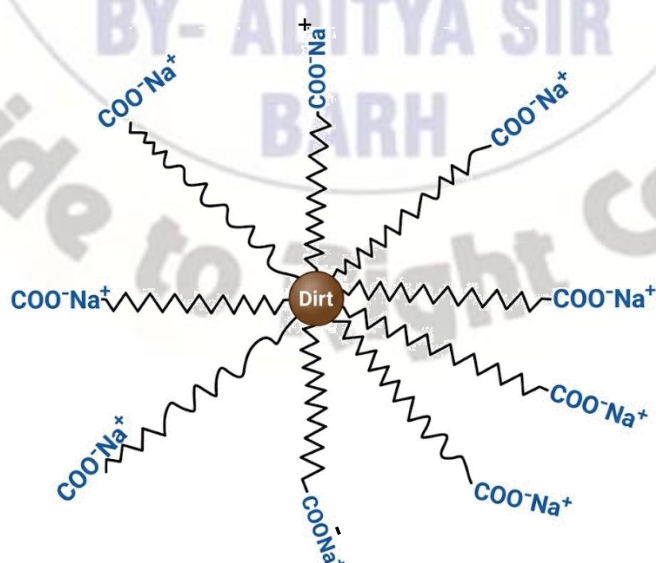
- Structure of soap molecule is like as tadpole structure. A soap molecule having two distinct part in which one part is a long hydrocarbon part and another is short ionic part containing $\text{COO}^- \text{Na}^+$.

Long hydrocarbon part is hydrophobic but ionic part is hydrophilic means water loving, hence ionic part is soluble in water but insoluble in oil and grease where as hydrocarbon part is insoluble in water but soluble in oil and grease.



Long hydrocarbon part OR Non – Ionic part / Non – Polar Part	Ionic part / Polar Part
Water hating / repelling	Water loving / Attracting
Dirt / Oil / Grease loving & Attracting	Oil / Dirt repelling / hating
Hydrophobic part	Hydrophilic part

• **Working rule of Soap**



- When soap is added with water its long hydrocarbon parts attached itself to the dirt. The dirt is thus invaloped in nonpolar ends of the soap micelles. The charge carboxylate of the soap molecules make the micelles soluble in the water.

Thus, the dirt is washed away with the soap and cloth is clean from dirt.

- **Working action of soap**

- The particles of smoothness and dust attached on the skin, cloth which are dissolved with soap and water.

- **The affects of soap on the skin and cloths by two types:**

- i) It combines with smoothness to formed emulsion which take away the particles of smoothness or dust.
- ii) It forms colloidal solution with water by which the particles of smoothness or dust are attached with soap. Hence skin and cloth is cleaned.

- **Behaviour of Soap with hard water**

- We know that hard water contains soluble calcium and magnesium salt of sulphate chloride and bicarbonate. When soap is added with hard water then the sodium salt of fatty acid present in the soap reacts with calcium and magnesium salt of hard water due to which insoluble calcium or magnesium salt of fatty acid is formed. i.e, isolated in the form of pericpitate. In this condition, cloth is more further dirt.

i.e, soluble salt of soap (Na^+) + Hard water $\rightarrow$ Insoluble salt (Soap). The bicarbonate of sodium remains rest in the water.

DETERGENT

- i) $\text{CH}_3 - (\text{CH}_2)_{11} - \text{C}_6\text{H}_4 \rightarrow \text{SO}_3^- \text{Na}^+$
Non – Ionic / └──────────┘ Ionic part (Sulphonate group)
Non – polar

Detergent: A detergent is the sodium salt of a long chain Alkyl hydrogen sulphate which has cleasing properties in water.

Name $\rightarrow$ Sodium n – dedocyl benzene sulphonate.

- ii) $\text{CH}_3 - (\text{CH}_2)_{10} - \text{CH}_2 - \text{SO}_3^- \text{Na}^+$
(Sodium n – dedocyl sulphate)

- iii) $\text{CH}_3 - (\text{CH}_2)_{10} - \text{CH}_2 - \text{SO}_3^- \text{Na}^+$
(Sodium turic Sulphate)

Structure:



- Detergent is also called soapless soaps because through they act like a soap in having cleaning properties.
- Detergent works well in the soft as well as hard water.
- The cleasing action of detergent is similar to that of a soap.

Special Note:

- Detergent is made first in Germany after second World War.

- **Behaviour of detergent with hard water**
- Detergent is better material than soap according to cleansing action because it easily dissolve in water and do not make insoluble calcium and magnesium salt. Thus, it makes more micelles with hard water.

Note:

- Detergent is totally non – biodegradable and it supports pollution in lakes and rivers.

- **Advantages of detergent over soap**

- i) It can be used every day with hard water whereas soap are not sustainable for use with hard water.
- ii) It has stronger cleansing action than soap.
- iii) It is more soluble in water than soap.
- iv) It can not be decomposed by micro – organism like bacteria.

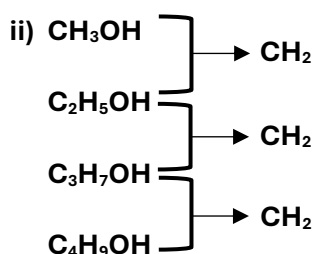
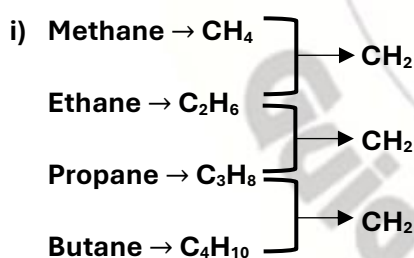
- **Difference between Soap and Detergent**

Soap	Detergent
Soaps are the sodium salt of the long chain carboxylic acids.	Detergent are the sodium salts of long chain alkyl hydrogen sulphate.
The ionic group in soap is COO^-Na^+	The ionic group in a detergent is SO_3^-Na^+
Soaps are not suitable for washing purpose when the water is hard.	Detergent can be used for washing ever when the water is hard.
Soaps are biodegradable.	Totally detergent are not biodegradable.
Soaps have relatively weak cleansing action.	Detergent have a strong cleansing action.

- **Homologous Series**

- A Series of organic compound having similar structures and chemical properties in which each membered differ from the next by CH_2 . The members of the series are called homologous series.

For example:-



Q. Give a test that can be used to differentiate between saturated and unsaturated hydrocarbon.

Ans For this test we take bromine water and put into unsaturated compounds. Then addition reaction take place Bromine Decolourised and Unsaturated Compound converted into saturated compounds in other hand unsaturated compounds decolourised the bromine water while saturated compound do not decolourised bromine water.

- **Define Catenation property of Carbon**

- The unique ability of carbon atoms to link with themselves to form long, stable chains, rings and complex structure (like single, double, or triple bonds) with other carbon atoms called catenation.

For example:-

- ✓ C_{60} – Buckminsterfullerene, Graphite & Diamond.

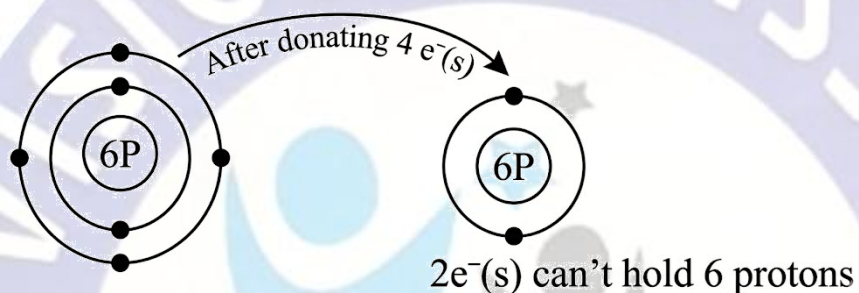
Q. Why carbon does not form 4 bonds?

OR

Why carbon does not gain or lose 4 electrons?

Ans

Case 1:



Case 2:

